



Orion-MC

Manual Airing Screens, BACnet
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Manual no. HH90042202



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1 Introduction

The Orion-MC is the communication interface between a horticulture computer (the main controller) and a variety of systems in the horticultural greenhouse (airing, screens).

The horticulture computer communicates with the Orion-MC using separate control signals. As an option, the communications between the horticulture computer and the Orion-MC can be routed via a data bus (BACnet).

The Orion-MC communicates with the IO modules (motor control cabinets) via a CAN Local data bus. The Orion-MC receives control commands from the main controller. The Orion-MC interprets these control commands in terms of commands for the relevant IO module (motor control cabinet). The Orion-MC also sends the main controller feedback on the position adopted by the controlled element.

The Orion-MC also offers a number of additional functions. The *open-roof greenhouse* option provides for the control of open-roof greenhouse windows, which can then be open and closed in the correct manner. The *two screens on one frame* option enables you to open and close screens on a cable frame without the screens sliding over each other.



Safety instructions and warnings

This product should be installed by an approved, certified installer in accordance with the relevant standards (BS, ANSI, ISO, for example).

The warranty is void if this product is installed in a switchbox other than by Hotraco Horti BV. The installation incorporating this product should be equipped with an appropriate safety system that is inspected at periodic intervals (at least once a day) to verify that it is working correctly.

Please note that when the product is included in a climate control system, the oxygen content in the climate controlled area must be guaranteed at all times, including during an emergency. To guarantee the oxygen content we recommend an emergency system, which must be periodically inspected for proper operation.

An emergency system can consist of:

- an emergency power generator;
- emergency opening systems;
- provision for manual operation.

We wish to draw your attention to the following:

- The Orion-MC computer is an electronic product. Consequently, you always need to take into consideration a possible malfunction.
The alarm contacts of the various computers should be connected to a central alarm system. Furthermore, Hotraco Horti BV recommends the installation of an additional independent alarm system (such as a min./max. thermostat).
Although Hotraco Horti BV has made every effort to ensure that adequate alarms are generated in the event of emergencies, Hotraco Horti BV cannot guarantee that all possible situations have been covered.
- We recommend that the computer remains powered up whenever possible. It is preferable to keep the computer powered up when not in use, since this will prevent condensation.
- Do not clean the computer with running water: the computer is splashproof, but not waterproof!
- This appliance is covered by the warranty and liability regulations laid down in documents including the Hotraco Horti general terms of sale governing the contract concluded for the delivery of this appliance.
- Use other than described in this manual is not permitted without express consent from Hotraco Horti BV.
- The Orion-MC is under no circumstances a safety component.
- The Orion-MC may only be operated and adjusted by persons who are familiar with the operating instructions provided in this manual.
- The person who operates or adjusts the Orion-MC must be aware of all potential dangers of the systems that the Orion-MC controls.
- A defective Orion-MC is hazardous! In the event of a fault, the systems in which the Orion-MC is used are to be taken out of service until the Orion-MC has been repaired.

- It is not permitted to connect external devices that do not have a SELV or Class II power supply (double insulated).
- Hotraco Horti wishes to point out that Machinery Directive 2006/42/EC is applicable when the Orion-MC is used to control moving systems.
- Due to the nature of the Orion-MC controllers, a controlled system can start at any time. Hotraco Horti would like to point out that this may result in hazardous situations and danger from moving parts when such a system is in operation. Therefore, people are not permitted to be in the immediate vicinity of moving parts. The moving parts must be fitted with guards and/or safety devices.
- Motors and actuators that are directly controlled via the Orion-MC may never be connected without a suitable motor protector and a lockable isolating switch.

End of life (WEEE)

The crossed-out wheeled bin indicates a product that, at the end of its life cycle, must be collected separately (Directive 2012/19/EU on waste electrical and electronic equipment - WEEE). Separate collection is a precondition to ensure specific treatment and recycling of WEEE to get the chosen level of protection of human health and the environment in the Union. The black bar below the crossed-out wheeled bin indicates that the product was put on the market on or after August 13, 2005



The European Union has systems in place for the separate collection of waste electrical and electronic products and batteries. Thus, return waste products, batteries and accumulators to the appropriate local authorities, or a local collection point/recycler. When you do not dispose of the product correctly, you may be fined.

2 Operation

2.1 Operating keys

The Orion-MC is equipped with a number of operating keys. These keys can be classified into the following groups:

Function keys: **F1** Time and date, groups, motors

F2

F3 No function

F4 Alarms

F5 Diagnostics

F6 Installation procedure

Navigation keys:  Move cursor

 Increase value / Previous line

 Decrease value / Next line

 Move cursor / Correction of numeric entry

OK Confirm value

 Next motor or motor group

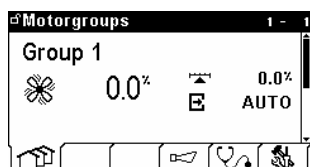
 Previous motor or motor group

2.2 The readout

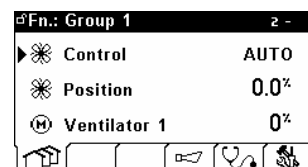
The display can show information in several ways. The section describes the main ways.

Main group

Press: **F1**



Settings



The *main group readout* displays the current data and/or the status of the controlled element. These data relate to the measured and calculated positions and can be fetched by pressing the **F1** key.

The data in the *main group readout* cannot be changed. You can change settings relating to the data displayed in the main group by pressing the **▶** key, which opens the settings readout. Press the **◀** key to return to the *main group readout*.

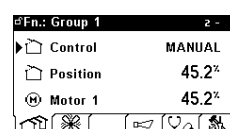
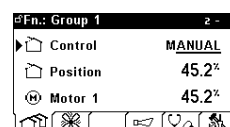
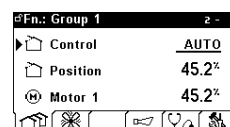
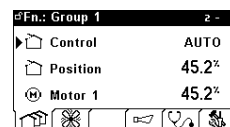
A list of the possible main groups and settings is enclosed in appendix 1. Data relating to controlled elements that have not been activated are skipped automatically and, consequently, cannot be fetched.

2.3 Entering and adjusting settings

Press the (**F1** or **F2**) function key to display the *main group screen*. Pressing the **▶** key opens the *settings screen*. You can fetch the required setting using the **▲▼** and **▼▲** keys.

Example Changing a group to manual operation mode:

1. Press **F1** or **F2**: the main group is displayed:
2. Press **▶**: the settings screen is displayed.
3. Press **▶**: the cursor is moved to the setting.
4. Press **▲**: the setting is changed.
5. Press **OK**: the changed setting has been confirmed.



2.4 System settings

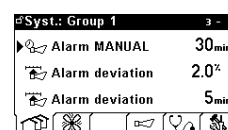
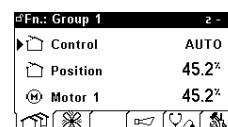
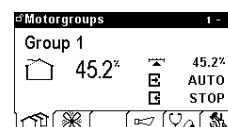
System settings refer to settings relating to the minor changes that can be made to the system. The basic settings have already been configured: the Orion-MC will carry out its control functions satisfactorily without any changes to the system settings.

However, you can change the system settings to tailor the Orion-MC precisely to the equipment that is to be controlled. We recommend that you make any changes in consultation with your installer.

The system settings are grouped by controlled element. The system settings for the relevant controlled element can be fetched and adjusted by opening the *settings* screen and pressing the **F6** key. The system settings that are not of relevance are not displayed.

Example System settings of a group:

1. Press **F1** or **F2**: the main group is displayed:
2. Press **▶**: the settings screen is displayed.
3. Press **F6**: the system settings are displayed. System settings are adjusted using the procedure used for the adjustment of settings, see chapter 2.3.

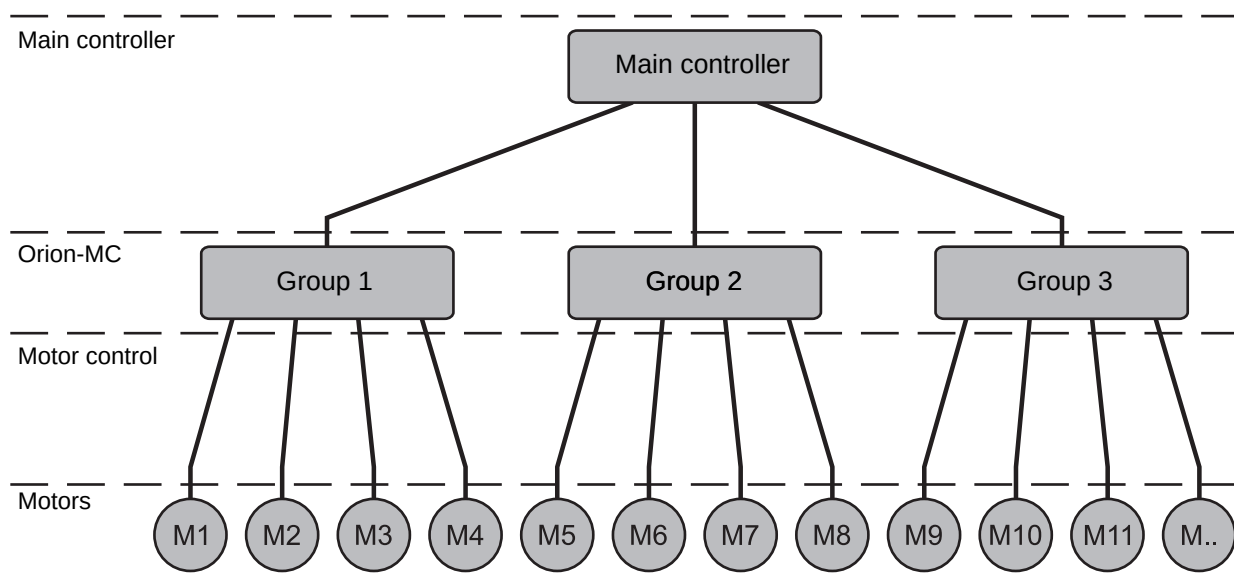


3 The Orion-MC's function

3.1 Control of the motor control cabinets (airing screens)

The building is equipped with a number of motors that can, for example, open and close the windows screens (M1, M2, etc., in the following diagram). These motors are connected to Hotraco Horti motor control cabinets that are in turn connected to the Orion-MC. See Appendix 1.1 and Appendix 1.2.

The motors are clustered into motor groups in the Orion-MC. Each motor group consists of one or more motors that are, in principle, controlled simultaneously and all adopt the same position. The main controller controls the motor groups and the Orion-MC ensures that the motors adopt the correct position.



You can use the Orion-MC to control all the motors in a group or, if you so wish, to control a single motor.

3.2 Virtual control

It is possible independent of the setpoint of the motor group, to shut of or switch on fans in this group. Up to maximal 12 digital inputs are available. All available fans can be assigned to one or more digital inputs.

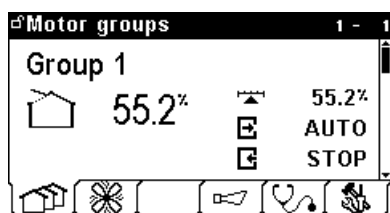
The horticulture computer serves as the controller and determines the required action for each group. The horticulture computer sends the relevant action to the Orion-MC. The Orion-MC simulates this action and sends the horticulture computer feedback on the simulated action (= virtual control). The horticulture computer interprets this feedback as confirmation that the required action has been carried out. At the same time the Orion-MC controls the IO modules (the motor control cabinets) by sending commands to ensure that the group carries out the required action. The relevant units send feedback on the positions to the Orion-MC. The action is stopped once the required position has been reached or an alarm is generated if the required position is not reached.

4 Airing settings

4.1 Window control

4.1.1 Group

Press **F1** to display the main **group** screen. You can use the **▲▼** keys to browse through the groups.



This screen displays the status of a **group** with window motors.

55.2% = The average of the measured positions of all window motors in the group. This can differ from the required position (setpoint).

▲ = The required position for the group, calculated by the main controller.

E = The status of the group in the Orion-MC:

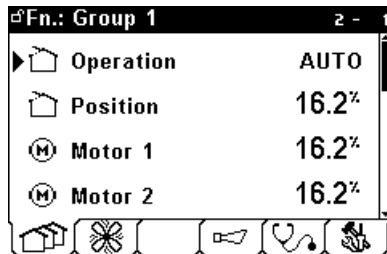
- AUTO** The group is set to automatic operation mode. The main controller determines the required position and the Orion-MC ensures that the window motors in the group adopt the required position.
- HAND** The group is set to hand operation mode. The group can be controlled via the settings in the Orion-MC.
- OFF** The group is switched off and is sent to the 0% position.

G = Action taken by the main controller (with digital control):

- OPEN** The main controller opens the windows in the group.
- CLOSE** The main controller closes the windows in the group.
- STOP** The main controller does not send any action.

4.1.2 Group settings

Press  in the main group to display the settings screen. You can use the  keys to browse through the groups.

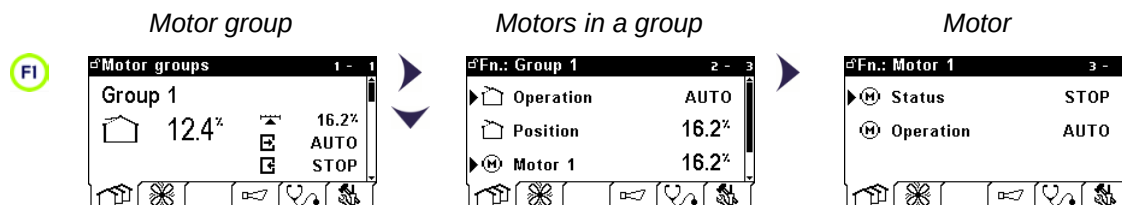


This screen displays the settings of the **group**.

-  = *Operation*: Change the status of the group:
 - AUTO* The group is controlled automatically; the main controller determines the required position and the Orion-MC ensures that the motors in the group adopt the correct position.
 - HAND* The group is controlled manually; the group can be controlled (using the *position* setting) from the Orion-MC.
 - OFF* The group is switched off: the group is sent to the 0% position.
- Note** the motor control cabinets to which the motors are connected must be set to automatic mode.
-  = *Position*: The required position for the group, calculated by the main controller. When the *Operation* of the group is set to *HAND* then you can set the position here.
-  = *Motor ...* : The current position of each motor. This example is of a group with two motors. Press  to view the **motor** settings (see chapter 4.1.3).

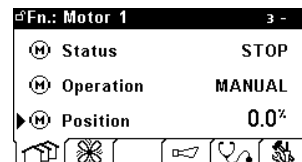
4.1.3 Motor settings

The settings of a single **motor** are fetched as follows.



You can use the  keys to browse through the motors. The screen displays the status of a single motor.

-  = *Status*:
- OPEN* The motor opens the window.
 - CLOSE* The motor closes the window.
 - STOP* The motor is stationary.
- Note** When this message is preceded by *EXT* then the motor is set to hand operation mode on the motor control cabinet.
-  = *Operation*: The operation mode of the motor can be set as follows:
- AUTO* The motor is set to automatic operation mode. The Orion-MC ensures that the motor adopts the correct position.
 - HAND* The motor is set to hand operation mode. An extra option that is displayed is used to set the motor position by hand.
 - OFF* The motor is switched off and is sent to the 0% position.
- Note** the motor control cabinet to which the motor is connected must be set to automatic mode.
-  = *Position*: This option is displayed when the motor is set to hand operation mode. Set the motor's setpoint here.



4.1.4 Group system settings

In the **group** settings screen, press  to fetch the system settings. Use the  keys to browse through the motor groups.

SYS Description

-  Alarm, hand
-  Alarm variance (min. variance before alarm is generated)
-  Alarm variance (min. wait time before alarm is generated)

Default	User
30 min	min
2%	%
5 min	min

Alarm, hand

When the **group** is still set to hand operation mode on the Orion-MC after the set time then the group's alarm contact is de-energised. This function can be disabled by setting *alarm, hand* at 0 minutes.

Alarm variance

You can configure the Orion-MC to generate an alarm in the event that a window is not at the required position after the set *wait time*.

4.1.5 Motor system settings

In the **motor** settings screen, press **F6** to fetch the system settings. Use the **◀▶** keys to browse through the motor groups.

SYS	Description
	Alarm, hand

Default	User
30 min	min

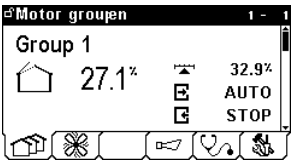
Alarm, hand

When the **motor** is still set to hand operation mode on the Orion-MC after the set time then the group's alarm contact is de-energised. This function can be disabled by setting *alarm, hand* at 0 minutes.

Note Other alarm contacts may also be de-energised, depending on the settings.

4.2 Open-roof greenhouse

The open-roof greenhouse function is configured during the installation procedure. The settings of the open-roof greenhouse are the same as the window control settings, see chapter 4.1. No extra user settings are available. More information about the open-roof greenhouse function is enclosed in Appendix 2.2.

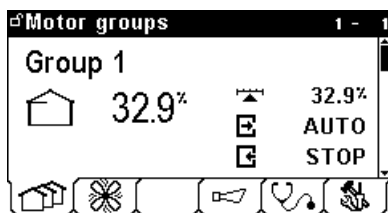


5 Screen settings

5.1 Screen control

5.1.1 Group

Press **F1** to display the main **group** screen. You can use the **▲▼** keys to browse through the groups.



This screen displays the status of a **group** with screen motors.

32.9% = The average of the measured positions of all screen motors in the group. This can differ from the required position (setpoint).

▲ = The required position for the group, calculated by the main controller.

E = The status of the group in the Orion-MC:

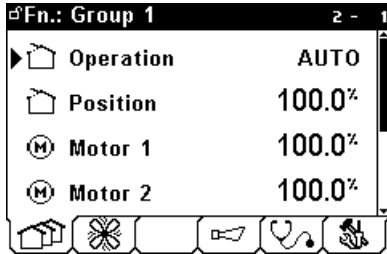
- AUTO** The group is set to automatic operation mode. The main controller determines the required position and the Orion-MC ensures that the screen motors in the group adopt the required position.
- HAND** The group is set to hand operation mode. The group can be controlled via the settings in the Orion-MC.
- OFF** The group is switched off and is sent to the 0% position.

G = Action taken by the main controller (with digital control):

- OPEN** The main controller issues a command for the screens (group) to be rolled up (opened).
- CLOSE** The main controller issues a command for the screens (group) to be rolled out (closed).
- STOP** The main controller does not send any action.

5.1.2 Group settings

Press  in the main group to display the settings screen. You can use the  keys to browse through the groups.

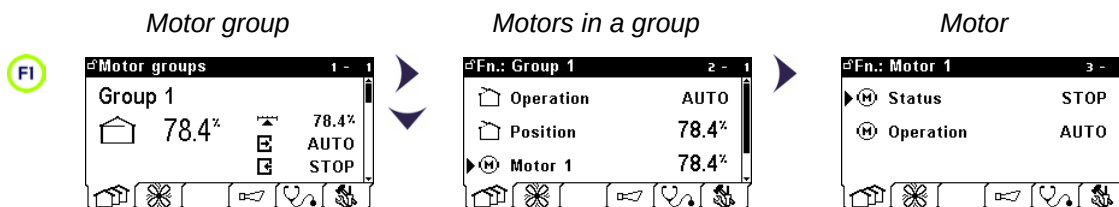


This screen displays the settings of the **group**.

-  = *Operation*: Change the status of the group:
 - AUTO* The group is controlled automatically; the main controller determines the required position and the Orion-MC ensures that the motors in the group adopt the correct position.
 - HAND* The group is controlled manually; the group can be controlled (using the *position* setting) from the Orion-MC.
 - OFF* The group is switched off: the group is sent to the 0% position.
- Note** the motor control cabinets to which the motors are connected must be set to automatic mode.
-  = *Position*: The required position for the group, calculated by the main controller. When the *Operation* of the group is set to *HAND* then you can set the position here.
-  = *Motor ...* : The current position of each motor. This example is of a group with two motors. Press  to view the **motor** settings (see chapter 4.1.3).

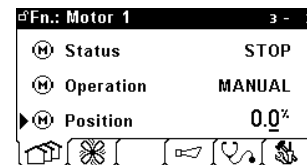
5.1.3 Motor settings

The settings of a single **motor** are fetched as follows.



You can use the  keys to browse through the motors. The screen displays the status of a single motor.

-  = *Status*:
- OPEN* The motor rolls the screen up (opened).
- CLOSE* The motor rolls the screen out (closed).
- STOP* The motor is stationary.
- Note** When this message is preceded by *EXT* then the motor is set to hand operation mode on the motor control cabinet.
-  = *Operation*: The operation mode of the motor can be set as follows:
- AUTO* The motor is set to automatic operation mode. The Orion-MC ensures that the motor adopts the correct position.
- HAND* The motor is set to hand operation mode. An extra option that is displayed is used to set the motor position by hand.
- OFF* The motor is switched off and is sent to the 0% position.
- Note** the motor control cabinet to which the motor is connected must be set to automatic mode.
-  = *Position*: This option is displayed when the motor is set to hand operation mode. Set the motor's setpoint here.



5.1.4 Group system settings

In the **group** settings screen, press  to fetch the system settings. Use the  keys to browse through the motor groups.

SYS Description

	Alarm, hand
	Alarm variance (min. variance before alarm is generated)
	Alarm variance (min. wait time before alarm is generated)
	Pulse zone
	Pulse width
	Cycle time

Default	User
30 min	min
2%	%
5 min	min
5%	%
1%	%
30 min	min

Alarm, hand

When the **group** is still set to hand operation mode on the Orion-MC after the set time then the group's alarm contact is de-energised. This function can be disabled by setting *alarm, hand* at 0 minutes.

Alarm variance

You can configure the Orion-MC to generate an alarm in the event that a screen is not at the required position after the set *wait time*.

Pulse system (pulse zone, pulse width and cycle time)

The pulse system can automatically close gaps caused by contraction and expansion. The pulse system is enabled during the installation procedure (see chapter 7.3). The pulse system can be used only when the screens are equipped with a slip system.

When the pulse system is enabled then you can adjust the following parameters in the system settings of a **group**.

Pulse zone ()

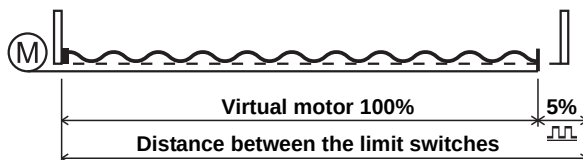
When the screen is opened or closed then the motor stops once the pulse zone is reached. The virtual motor is then in the minimum or maximum position (so that the main controller receives feedback that the screen is in the correct position).

Pulse width () and Cycle time ()

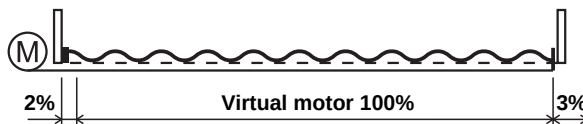
Once the screen is at the beginning of the pulse zone then the screen receives a command to close or open further at intervals (the cycle time) by a distance equal to the pulse width. This cycle stops at the end of the pulse zone.

Example

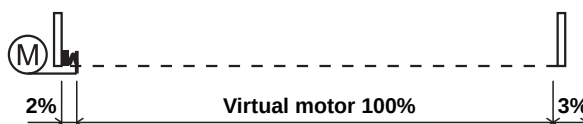
Pulse zone  = 5%; Pulse width  = 0.2%; Cycle time  = 20 minutes. The main controller closes the screen fully to 100%. The Orion-MC stops the drive motor at the beginning of the pulse zone.



Once every 20 minutes (the cycle time) the screen is sent a command to move a further 0.2%: the virtual motor moves the same distance. After 200 minutes (10x 0.2%) the virtual motor has moved a further 2%. A pulse zone of 2% has then developed at the other end of the screen. Once the virtual motor has moved 5% (in this example, after 8 hours and 20 minutes) then the pulse system is stopped. The pulse zone at the other end is then 5%.



When the main controller opens the screen fully to 0% 200 minutes later then the Orion-MC stops the drive motor 2% before the limit point (the beginning of the pulse zone). The virtual motor is then at 0%. The screen is now opened further by 0.2% every 20 minutes (cycle time); the virtual motor moves the same distance.



5.1.5 Motor system settings

In the **motor** settings screen, press **F6** to fetch the system settings. Use the **⟷** keys to browse through the motor groups.

SYS Description

 Alarm, hand

Default	User
30 min	min

Alarm, hand

When the **motor** is still set to hand operation mode on the Orion-MC after the set time then the group's alarm contact is de-energised. This function can be disabled by setting *alarm, hand* at 0 minutes.

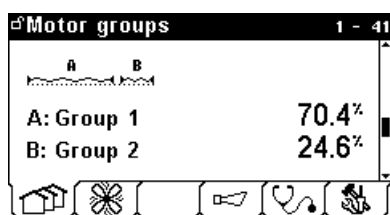
Note Other alarm contacts may also be de-energised, depending on the settings.

5.2 Two opposing screens on one cable frame

The Orion-MC is equipped with a function that controls the screens in a manner which ensures that they cannot move over each other. This system can be configured during the installation procedure (see chapter 8.6). More information is enclosed in Appendix 2.3.

5.2.1 Main group screen, 'Two opposing screens on one cable frame'

Press **F1** and use the **▲▼** keys to move to the main group screen. A number of combinations can be compiled.



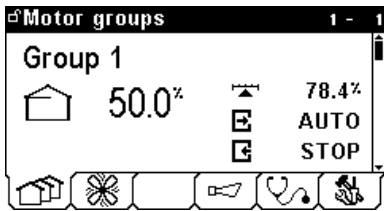
The main group screen displays the Orion-MC setting for the groups. In this example group 1 has been controlled to 70.4% and group 2 has been controlled to 24.6%. For safety purposes a 5% gap is specified between the groups. The total is 100%. The size of the gap is specified by a setting (see chapter 5.2.2).



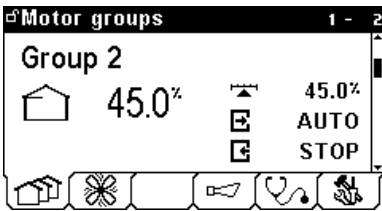
The controlled percentages do not need to be equal to the required position specified by the main controller: the Orion-MC adjusts the actual controlled position to avoid an overlap.

Example

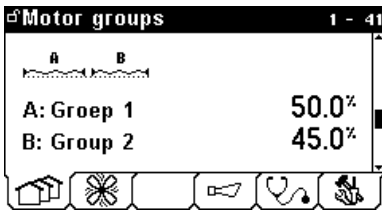
Required position of group 1



Required position of group 2



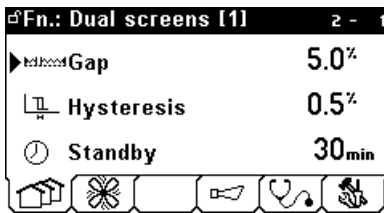
Pos. controlled by the Orion-MC



When no master group has been specified then the group that has been sent a close command last is assigned priority. In the example group 2 was the group sent a close command last, as a result of which the required position is in agreement with the controlled position. When one of the groups is the master group then the master group is always set to the required position. More information is enclosed in Appendix 2.3.

5.2.2 Settings, 'Two opposing screens on one cable frame'

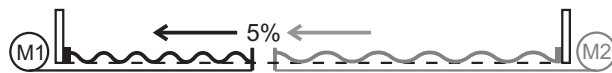
Press  in the main group to display the settings screen. You can use the   keys to browse through the groups.



-  = **Gap**: The gap between the two screens.
-  = **Hysteresis**: This percentage determines the extent to which one screen may catch up with or lead the other screen at the time that one screen sends the other screen back.

Example gap = 5%, hysteresis = 1%.

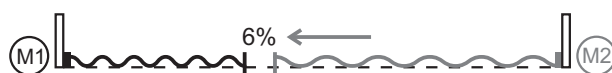
Group 2 has approached group 1, as a result of which group 1 is sent a command to move back.



When the gap becomes smaller than 4% (5% gap – 1% hysteresis) while both screens are moving then group 2 will stop until the gap increases to 5%.



When the gap becomes larger than 6% (5% gap + 1% hysteresis) while both screens are moving then group 1 will stop until the gap decreases to 5%.



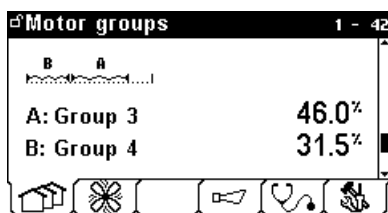
-  = **Standby**: The period the Orion-MC must wait for the gap to be closed. The position of neither screen may change during this interval. This setting is available solely when the sleep state option has been enabled during the installation procedure.

5.3 Two consecutive screens on one cable frame

The Orion-MC is equipped with a function that controls the screens in a manner which ensures that they cannot move over each other. This system can be configured during the installation procedure (see chapter 8.6). More information is enclosed in Appendix 2.3.

5.3.1 Main group screen, two consecutive screens on one cable frame'

Press  and use the   keys to move to the main group screen. A number of combinations can be compiled.

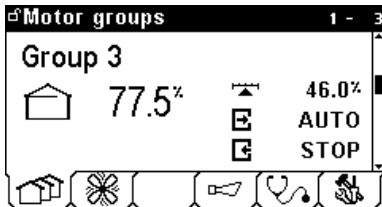


The main group screen displays the Orion-MC setting for the groups. In this example group 3 has been controlled to 46.0 % and group 4 has been controlled to 31.5 %.

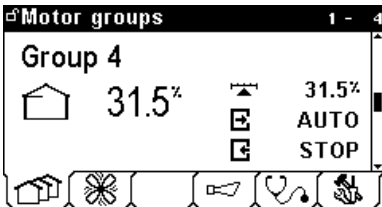
The controlled percentages do not be equal to the required position specified by the main controller: The Orion-MC adjust the actual controlled position to ensure that the screen sections are not squashed.

Example

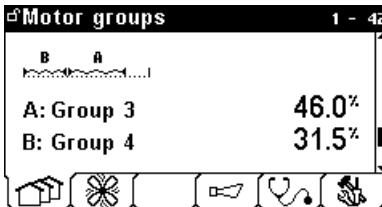
Required position of group 3



Required position of group 4



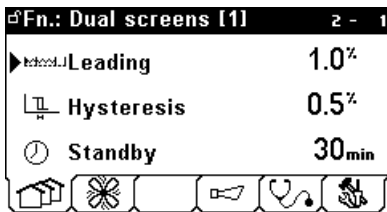
Pos. controlled by the Orion-MC



When no master group has been specified then the group that has been sent a close command last is assigned priority. In the example group 4 was the group sent a close command last, as a result of which the required position is in agreement with the controlled position. When one of the groups is the master group then the master group is always set to the required position. More information is enclosed in Appendix 2.3.

5.3.2 Settings, 'Two consecutive screens on one cable frame'

Press  in the main group to display the settings screen. You can use the   keys to browse through the groups.



-  = **Lead**: The minimum distance to be maintained between the screen sections when the screens are moving synchronously.
-  = **Hysteresis**: This percentage determines the extent to which one screen may catch up with or lead the other screen at the time the screens are moving synchronously.

Example *lead* = 5%, *hysteresis* = 1%.

Group 1 and group 2 are moving synchronously.



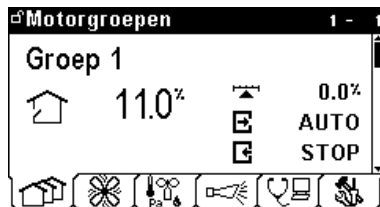
When the lead becomes smaller than 4% (5% gap – 1% *hysteresis*) while both screens are moving then group 2 will stop until the lead increases to 5%.



When the lead becomes larger than 6% (5% gap + 1% *hysteresis*) while both screens are moving then group 2 will stop until the lead decreases to 5%.



-  = **Standby**: The period the Orion-MC must wait for the lead to be closed. The position of neither screen may change during this interval. This setting is available solely when the sleep state option has been enabled during the installation procedure.



6 Alarms

6.1 Alarm messages

6.1.1 Alarm types

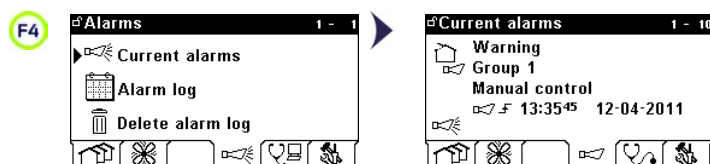
The Orion-MC uses the following types of alarms (F4).

-  = Soft alarm. The alarm contacts on the IO boards remain energised.
-  = Several soft alarms.
-  = Hard alarm. The alarm contacts on the IO boards are de-energised.
-  = Several alarms, including at least one hard alarm.

Each IO board has a separate alarm contact that is de-energised in the event of a communication failure, power failure and a **hard alarm**. The alarm contact activates, for example, a claxon, revolving lamp and/or telephone dialler. Your installer will configure the necessary settings for additional alarm contacts during the installation procedure. The following options are available:

- Each group has a separate alarm contact that is de-energised in the event of a **hard alarm** relating to the relevant group.
- A separate alarm contact that is de-energised on each **soft alarm**.

6.1.2 Alarm display



6.1.3 Resolve the alarms

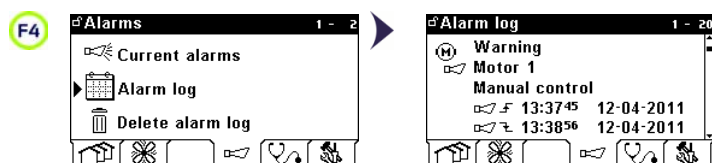
Use the following procedure in the event of an alarm:

- Press F4 to view which alarm has been generated. When there are several alarms then you can use the   keys to browse through the alarms.

2. Most alarms can be converted into another type of alarm. You do so by pressing **OK** for the alarm you wish to convert and then using the $\nabla/\blacktriangle$ keys to change the type of alarm.
 -  = **Soft alarm:** The alarm has yet to be resolved, but the alarm contacts on the IO boards are re-energised. The alarm contact of the group remains de-energised. The separate alarm contact for soft alarm is de-energised.
 -  = **Suppressed soft alarm:** The alarm has not yet been resolved but will no longer be generated. The alarm contact of the group remains de-energised. When the alarm contact for soft alarm is de-energised then this contact is re-energised.
 -  = **Hard alarm:** When the alarm contact for the group has been de-energised then this remains de-energised. When the alarm contact for soft alarm is de-energised then this contact is re-energised. The alarm contacts on the IO boards are de-energised.
 -  = **Deleting an alarm because it has been resolved:** If the alarm has been resolved then you can delete the alarm to restore the control.
3. Investigate the cause of the alarm.
4. Solve the problem in a **safe manner**.
5. Many alarm messages will be cancelled automatically once the problem has been resolved. When necessary, alarms will be cancelled only once you have deleted the message with the  command. When an alarm cannot be cancelled then there is still a problem.
6. Once all alarms have been cancelled *No current Alarms* is displayed in the screen.

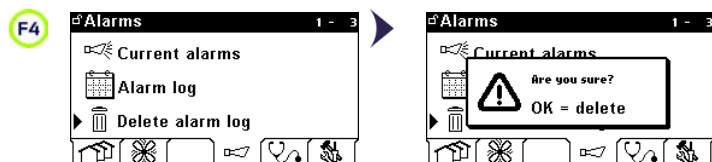
6.1.4 Alarm log

Once an alarm has been resolved then the alarm message is saved in *Alarm log*. The time at which the alarm was resolved is added to the last line. You can use the keys to browse through the alarm messages.



6.1.5 Deleting alarms

You can use *Delete alarm log* to delete the alarms.



6.2 Alarm messages from the IO modules

6.2.1 Motor control cabinets, general

The following messages can be sent from the motor control cabinet (H1MC, H2MC and CAN-IO-EKU) to the Orion-MC.

Description	Type	Cause
 Motor 'xx' Emergency limit switch'.	Alarm	Motor number 'xx' has moved beyond the regular limit switch and has ultimately been stopped by the emergency limit switch.
 Motor 'xx' Thermal cut-out closed or thermal cut-out open.	Alarm	Motor number 'xx' draws too much current and the thermal cut-out in the motor control cabinet has been activated.
 Motor 'xx' Break input.	Alarm	The break protection of motor 'xx' in the motor control cabinet has broken.
 Motor 'xx' Motor speed too low'	Alarm	The speed of motor 'xx' is too low. The encoder pulse frequency is too low (minimum of 0.5 Hz).
 Motor 'xx' Error signal encoder.	Alarm	The signal from the encoder is not correct. The encoder is defective or not connected properly.
 Motor 'xx' No feedback.	Alarm	The output of motor number 'xx' is being increased or decreased manually without feedback. Automatic operation is no longer possible.
 Motor 'xx' Position differs.	Alarm	The encoder position of motor number 'xx' differs from that in the limit position by more than 1.5%.
 Motor 'xx' Position differs.	Warning	The encoder position of motor number 'xx' differs from that in the limit position by more than 1%.
 Motor 'xx' Too few pulses.	Alarm	The encoder of motor number 'xx' generates too few pulses. The read-in has been rejected. The motor must be read in again.
 Motor 'xx' Too many pulses.	Alarm	The encoder of motor 'xx' generates too many pulses. The read-in has been rejected. The motor must be read in again (max 50 Hz).
 Motor 'xx' Installation mode.	Warning	The motor control cabinet for motor number 'xx' is in the installation procedure.
 Motor 'xx' Direction unknown.	Alarm	The direction of movement of motor number 'xx' is not correct (read-in procedure). The motor must be read in again.
 Motor 'xx' Encode error. A or B signal.	Alarm	The board is not receiving an A or B signal from the encoder.

Description	Type	Cause
 Motor 'xx' not configured.	Alarm	Motor number 'xx' has not been read in.
 Motor number 'xx' Error signal encoder.	Alarm	Pulses are being counted while motor number 'xx' is stationary. This can be due to interference signals affecting the encoder.
 Motor 'xx' Dual screen not possible	Alarm	<i>Dual screen guard</i> in the motor control cabinet cannot be adjusted because the condition is not met.
 Motor 'xx' Sensor hi-speed	Warning	The sensor monitoring the reduce speed position has been activated before the speed of the motor has been reduced (frequency controller).
 Motor 'xx' Frequency controller.	Alarm	The frequency controller of motor number 'xx' has reported a malfunction.

6.2.2 Motor control cabinets H1MC

The following messages can be sent by a H1MC motor control cabinet with slave motors to the Orion-MC.

Description	Type	Cause
 Motor number 'xx' speed not equal.	Alarm	The synchronised motors have different speeds.
 Motor 'xx' Sync guard active.	Alarm	The asynchronous movement protection has been activated. Check the drive.
 Motor 'xx' Limit switch not reached.	Alarm	One H1MC has not reached its limit switch whilst the linked H1MCs have reached their limit switches.
 Motor 'xx' Direction of rotation incorrect.	Alarm	The direction of rotation indicated by the encoder of a slave motor differs from that of the master motor.
 Motor 'xx' More than master.	Alarm	More than one H1MC motor in a driveline has been configured as a master motor.
 Motor 'xx' Limit switches not equal.	Alarm	The end positions of the synchronised motors differ by more than 1%.

6.3 Orion-MC Alarms

Description	Type	Cause
 Motor 'xx' Hand mode.	Alarm/Warning	Motor number 'xx' is set to hand-operation mode (in the Orion-MC or on the motor control cabinet). The alarm contact will be de-energised after a while (see system settings).
 Group 'xx' Manual mode.	Alarm/Warning	The group number 'xx' is set to manual-operation mode in the Orion-MC. The alarm contact will be de-energised after a while (see system settings).
 Motor 'xx' No motor active.	Warning	The motors of group number 'xx' are not active. These motors have been switched off or set to hand-operation mode in the Orion-MC.
 Motor 'xx' Position not reached.	Alarm	Motor number 'xx' has not reached the required position (see system settings).
 Motor 'xx' Link not configured.	Alarm	Motor number 'xx' has been configured for operation together with another motor but has not been linked to the relevant motor (installation procedure: open-roof greenhouse and/or two screens on one frame).

6.4 System alarms

6.4.1 General alarms

Description	Type	Cause
 Orion switched off	Warning	The software has switched off the computer at the specified time.

The computer has been switched off and gives an alarm. Press any key to suppress the alarm and to convert the alarm into a warning. No alarms from the controllers are shown while the computer is switched off.

Description	Type	Cause
 System restart	Warning	The computer has been restarted at the specified time.
 Settings deleted	Alarm	The settings have been deleted at the specified time (Installation procedure).

Description	Type	Cause
 Options deleted	Alarm	The options have been deleted at the specified time (Installation procedure).
 Alarm deleted	Warning	The alarms have been deleted at the specified time (see chapter 6.1).
 Options restored	Alarm	Someone has restored the options.
 Settings restored	Alarm	Someone has restored the settings.
 Options + settings restored	Alarm	Someone has restored the options and settings.
 Options (+ settings) not restored	Alarm	The options and settings that have been restored are not correct.

6.4.2 Communication alarms

Description	Type	Cause
 Board IO-xx-xx [xx] not found.	Alarm	The Orion has no connection with the relevant IO board. Possible causes: • Communication interrupted. • Broken contact. • CAN bus terminator defect.
 No alarm contact.	Alarm	No alarm contact assigned. The IO boards are not configured correctly (Installation procedure).
 Motor 'xx' not found.	Alarm	The motor control cabinet has a communication problem.
 Motor 'xx' with Slave 'xx' not found.	Alarm	There is a problem with the CAN-Sync connection of the H1MC.

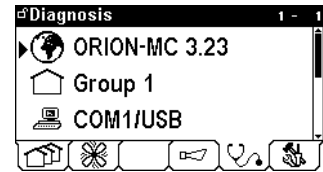
6.4.3 System alarms (internal)

The Orion-MC tests a number of internal functions that are directly applicable to the computer. On detecting an error the computer generates a system alarm displaying one of the following messages. Always warn your installer in the event of a system alarm!

Description	Type	Cause
 Options corrupted	Alarm	System setup no longer correct.
 Settings corrupted	Alarm	System settings are no longer correct.
 Minimum/maximum value corrupted	Warning	The stored minimum and maximum values are no longer correct.
 I ² C error	Alarm	Internal communication problem in the Orion.
 EEPROM error	Alarm	Internal communication problem with the EEPROM settings and options.
 RTC error	Warning	Problem with the real-time clock (program flow).
 TIMER error	Warning	Internal clock is not working correctly.
 HTRAP error	Warning	Internal memory is not working correctly.
 PLL error	Warning	PLL circuit is not working correctly.
 IO board ADC error	Warning	Analog/Digital converter not working properly.

7 Diagnostics

Press the **F5** diagnostics keys to get access to extra functions. You can use these functions to access information about the system elements. This information is clustered by controlled group. The diagnostic functions are displayed by pressing the **►** key for the relevant element.



You can also fetch information about the various communication lines. Only the installed communication lines are displayed (see chapter 7.4).

You can also use the diagnostics screen to read the data of a SD card or write data to a SD card. More information is given in chapter 7.5

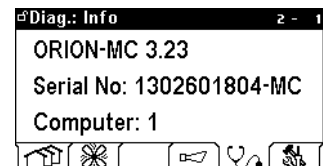
7.1 Software version, serial number and module

Software version

The diagnostics opening screen displays the software's version (in this example Orion-MC 4.18). Since the majority of this computer's functions are determined by the software it is important that the number in front of the decimal point (in this example 4.18) is in agreement with the number of this manual (see the version number on the front page + footer texts).

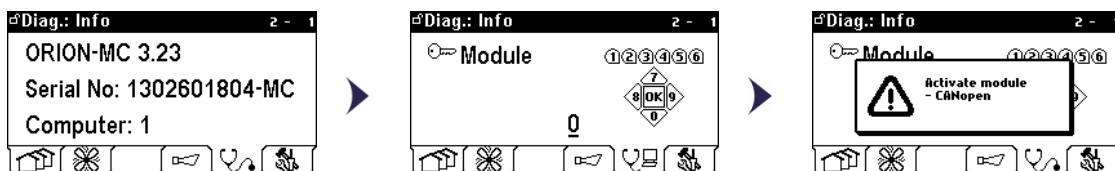
Serial number

Each Orion is assigned a unique serial number. When the cursor is on *Orion-MC*, pressing the **►** key displays the screen with the serial number. This can be applicable to a software update.



Module

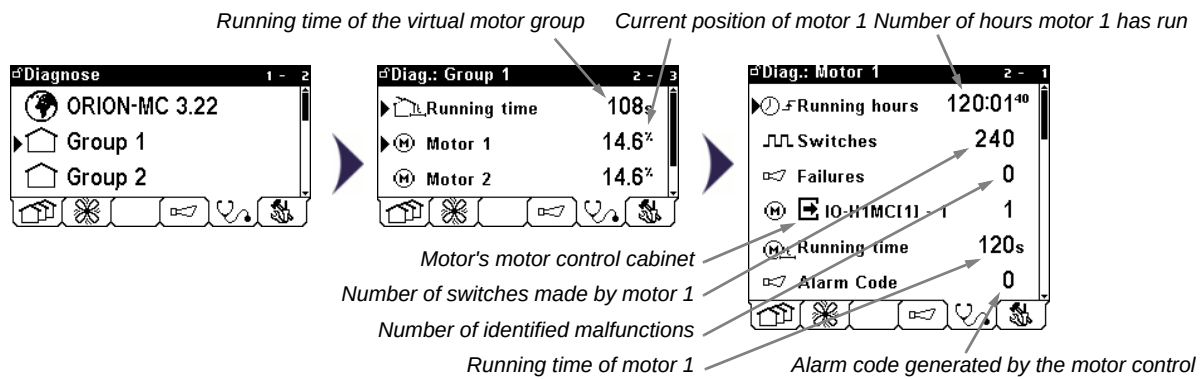
You can change or add a module (for example, CANopen communication) by entering a code under *Module* (**C↵**).



Note You can request the code from your supplier.

7.2 Window control

Pressing on the **►** key when the cursor is on *Group* fetches diagnostics information for the group with window motors. You can use the **▲▼** keys to select one of the motors in the group. You cannot make any changes.

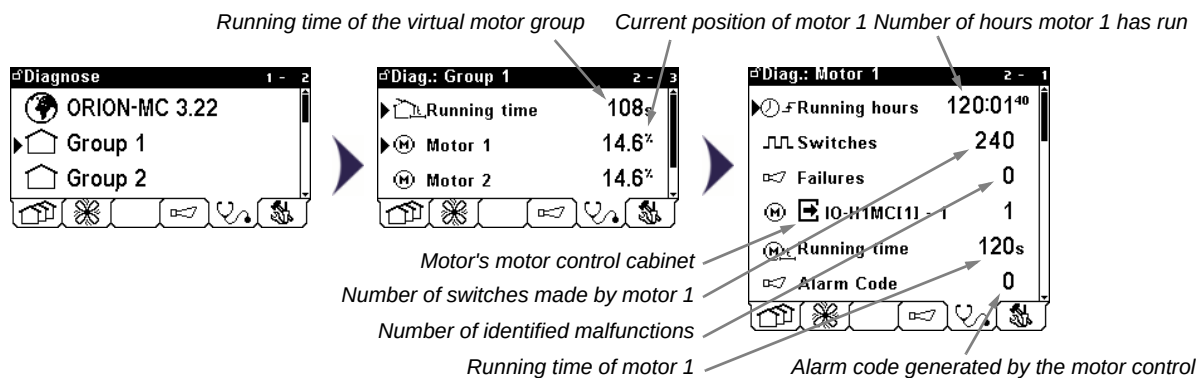


The running time of the group (virtual motor) is equal to 90% of the running time of the fastest motor in the group.

Example When the fastest motor has a running time of 120 seconds then the running time of the group is 108 seconds (90% of 120 seconds).

7.3 Screen control

Pressing on the key when the cursor is on *Group* fetches diagnostics information for the group with screen motors. You can use the keys to select one of the motors in the group. You cannot make any changes.



The running time of the group (virtual motor) is equal to 90% of the running time of the fastest motor in the group.

Example When the fastest motor has a running time of 120 seconds then the running time of the group is 108 seconds (90% of 120 seconds).

7.4 Communication

7.4.1 COM1/USB and COM2

Diag.: COM1/USB 38.4kBd

TX	94	449
RX	5	62
TX+RX	99	463

Diag.: COM2 38.4kBd

TX	114	243
RX	0	0
TX+RX	113	243

AT ☐

Modem DCD OK ☐

Modem Init ☐

Communication information

Send AT command to the Modem

Test DCD signal

Send initialisation string to the Modem

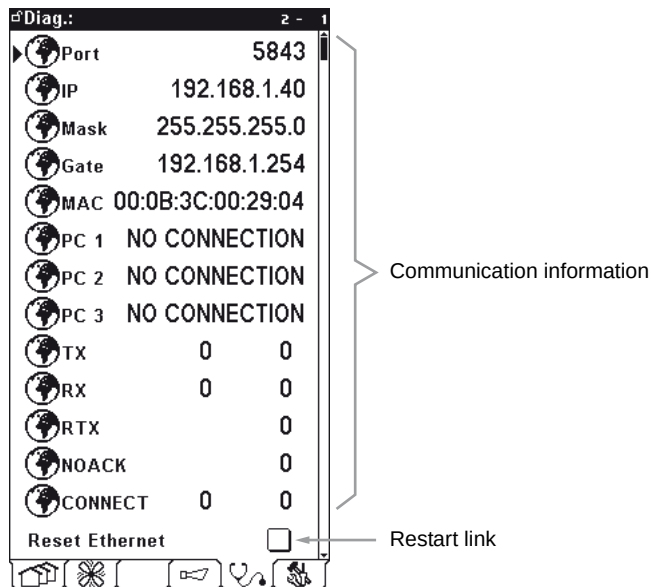
7.4.2 CAN Local (IO boards)

Diag.: CAN LOCAL 2 - 1

TXOK	1	225
RXOK	1	241
TXOK+RXOK	0	459
BOFF		0
EWRN		0
Stuff error		0
Form error		0
Ack error		0
Bit1 error		0
Bit0 busoff error		0
Bit0 normal error		0
Crc error		0

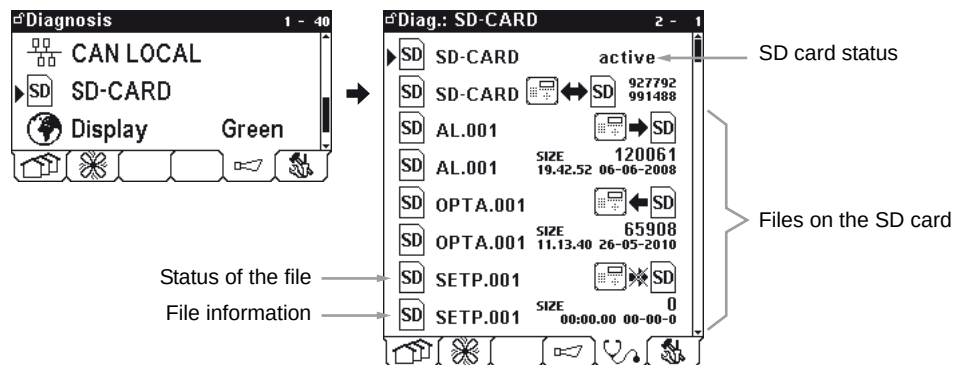
Communication information

7.4.3 Ethernet (BACnet)



7.5 SD card

7.5.1 SD card settings



Changing the SD card status

Press the **▶** key when the cursor is on *SD CARD*. Change the status with the **▲▼** keys. Confirm with the **OK** key. You can assign the following statuses to the SD card:

Remove: The SD card can be removed safely from the Orion-MC.

Ask remove: Data is still being written to the card, the SD card cannot be removed yet.

Current: The SD card can be used to save or restore data.

Changing the status of the file

Press the  key when the cursor is on the file. Change the status with the   keys. Confirm with the **OK** key. You can assign the following statuses to the SD card:

-  No action takes place.
-  The data are stored on the SD card. With log data the data are supplemented continually and the  symbol is always displayed. With one-off data (such as options) the symbol changes back to  once the data has been saved.
-  The data are read out from the SD card and send to the Orion (for example, options). Once the read-out has been completed the symbol changes back to .

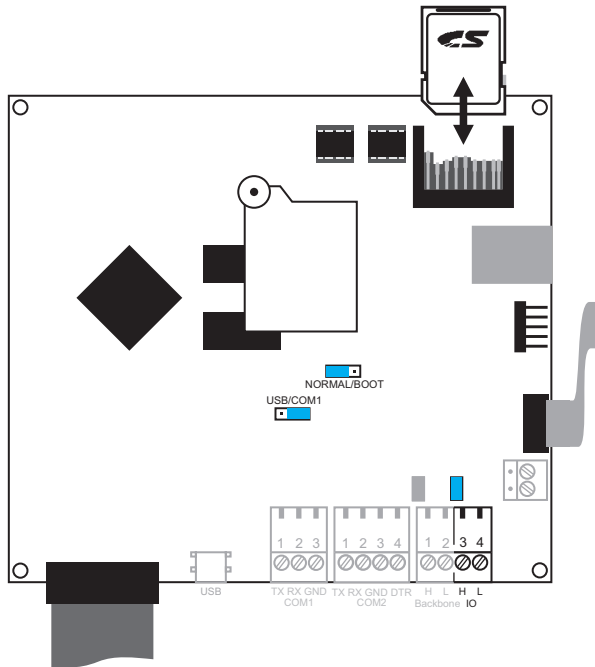
Data on the SD card

Alarm list (AL), options (OPT) and setpoints (SETP).

7.5.2 SD card location

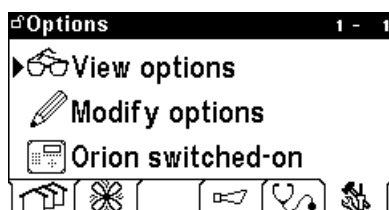
The SD card is in the case of the Orion-MC. Open the case to access the SD card. The SD card is located on the CPU board. Carefully slide the SD card into or out of the holder. Only FAT-16 or FAT-32 format SD cards can be used: SDHC cards cannot be used.

Note Make sure that the status of the SD card is set to *Remove* (see chapter 7.5.1).

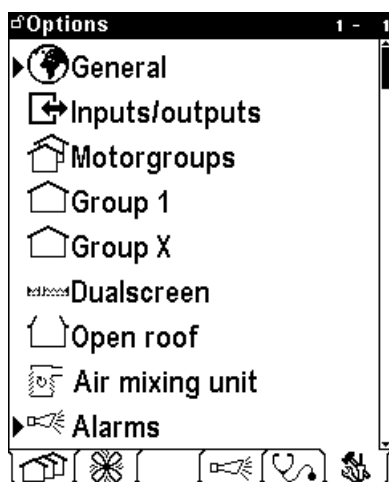


8 Installation procedure

Once the Orion-MC has been connected in accordance with the connection diagrams (see chapter 9) then you can enable the required options. Pressing the **F6** key when the main group screen is displayed opens the following screen:



This screen can be used to view and/or change the options and switch the Orion-MC on or off with the software. Moving the cursor to *View options* or *Change options* and then pressing **►** opens the option groups, provided that these are not secured with an access code. You always have access to *View options*, even when changes cannot be made without an access code. The access code is set in the *General* group (see chapter 8.1). The option groups that can be configured during the installation procedure are:



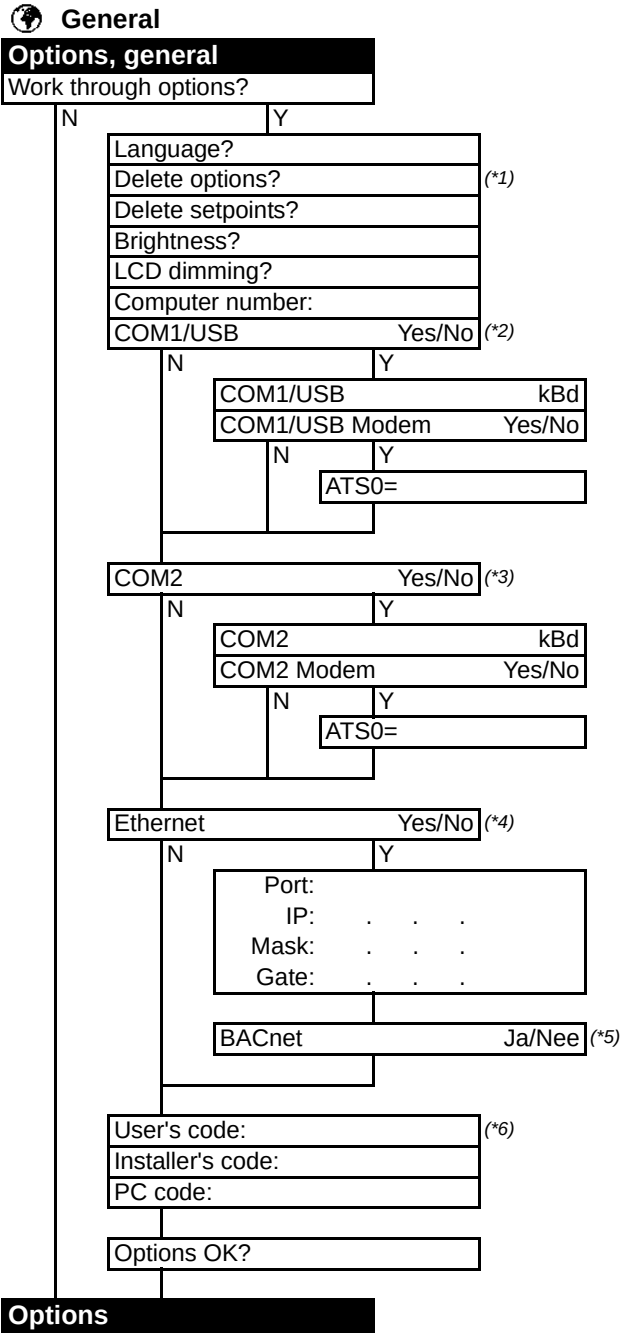
Note Your screen may differ from the screen shown above: solely the option groups of the available modules are displayed!

You can navigate through the various option groups with **▼** or **▲** and then **►**. In the options screen you can use **▼** and **▲** to navigate through all questions: press **►** to change the option settings.

Fill in the configuration of each option group in the charts shown on the following pages.

8.1 General options

The general options include the language, communication settings and access codes.



- (*1) Set the required language. If so required, you can delete existing options and setpoints. You can adjust the brightness of the screen. Activating LDC dimming will automatically dim the screen after 1 minute of no activity.
- (*2) Enable the COM1 or USB port here. In addition, set the communication speed (default: 38.4 kBd). Under Answer (ATS0=), set the number of rings before the modem answers.
- (*3) Enable the COM2 port here. In addition, set the communication speed (default: 38.4 kBd). Under Answer (ATS0=), set the number of rings before the modem answers.
- (*4) Enter the network settings here.
Port = the port number.
IP = the IP address of the Orion
Mask = Definition of the local network.
Gate = IP address of the router.
Use the ◀ and ▶ keys to browse through the settings. You can use the ▼ and ▲ keys to change the figures.
- (*5) Enable BACnet if your system uses BACnet.
- (*6) A password consists of digits. When the password is to be entered the keys of the Orion are assigned the numerical values displayed on the screen (F1=1, F2=2, ▲=7, etc.). Confirm the entry with **OK**. Under Repeat, enter your selected password again.
Passwords may not begin with 0 and must consist of at least 1 and at most 4 digits.
Entering 0 (▼) disables the password function.
The user's access code can be used to disable access to the control settings and installation procedure. The settings are automatically disabled once you have not pressed any key for 10 minutes.
The installer's access code can be used to secure access to the installation procedure.
The PC code secures logging in via PC software.

8.2 IO module settings

You use the IO modules options to configure the inputs and outputs.

Options: IO-08-09

Work through options?

N Y

Number of IO module: 1..8

Digital inputs: (*7)

Analog outputs: (*8)

Digital outputs: (*9)

Alarm contact Yes/No (*10)

Copy options to IO number: (*11)

Options OK?

Y

Options: IO

- (*7) The current value of the input is shown in the display (ON or OFF).
- (*8) Press **OK** to configure the output. You can select from a 0-5 V or 0-20 mA signal (▼▲). Press **OK** to set the minimum and maximum of the output.
- (*9) Press **OK** to test the output with the ▼▲ keys.
- (*10) Yes = the alarm contact of this module is de-energised in the event of a hard alarm. No = the alarm contact is de-energised only when the CAN-Local communication is broken or in the event of a power failure.
- (*11) The options of identical IO modules are often the same. You can use *Copy options* to copy the IO module's settings. Enter the CAN Local address of the new IO module and confirm with **OK**. Confirm again with **OK**. The new IO module will then be created with the same options. You can then continue immediately with the options of the newly created IO module.

Options: H1MC/H2MC/EKU (*12)

Work through options?

N Y

Number of IO module: (*13)

Motor control: (*14)

Alarm contact Yes/No (*15)

Options OK?

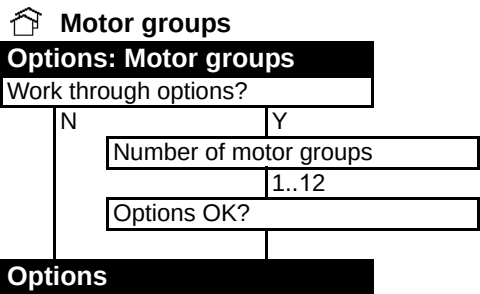
Y

Options: IO

- (*12) The options for the ECU, H2MC and H1MC are virtually identical.
- (*13) Enter the CAN Local address configured on the IO module here.
- (*14) Enter the correct type of motor.
Window motor (100% = open, 0% = closed)
Screen motor (100% = open, 0% = closed)
You can also view the current setting for the position here.
- (*15) Yes = the alarm contact of this module is de-energised in the event of a hard alarm. No = the alarm contact is de-energised only when the CAN-Local communication is broken or in the event of a power failure.

8.3 Motor groups

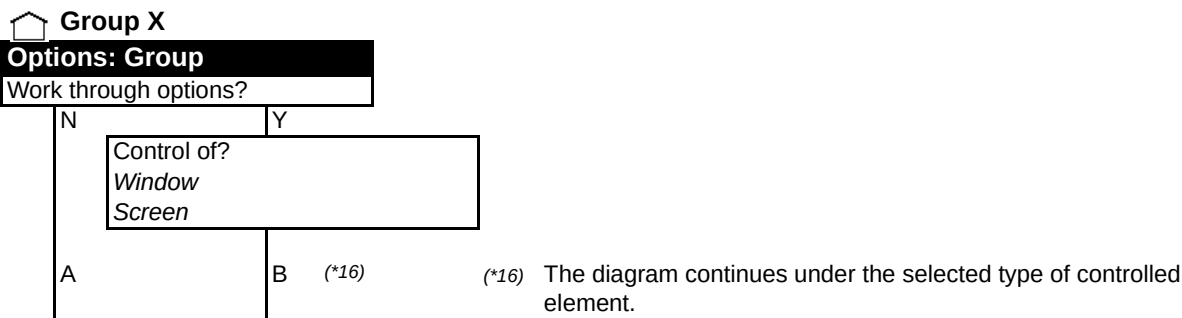
A motor group consists of motors (fans that are controlled simultaneously).
See also Appendix 2. Configure the number of motor groups here.



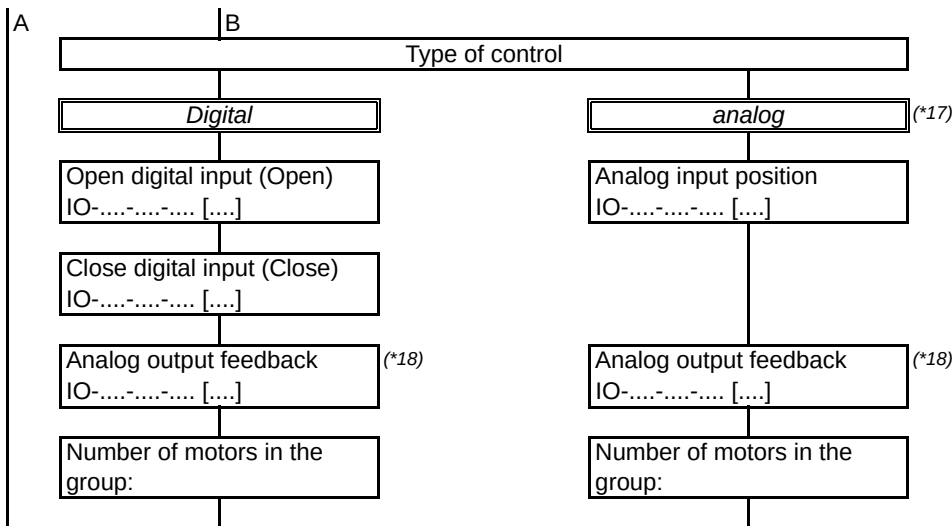
8.4 Group settings

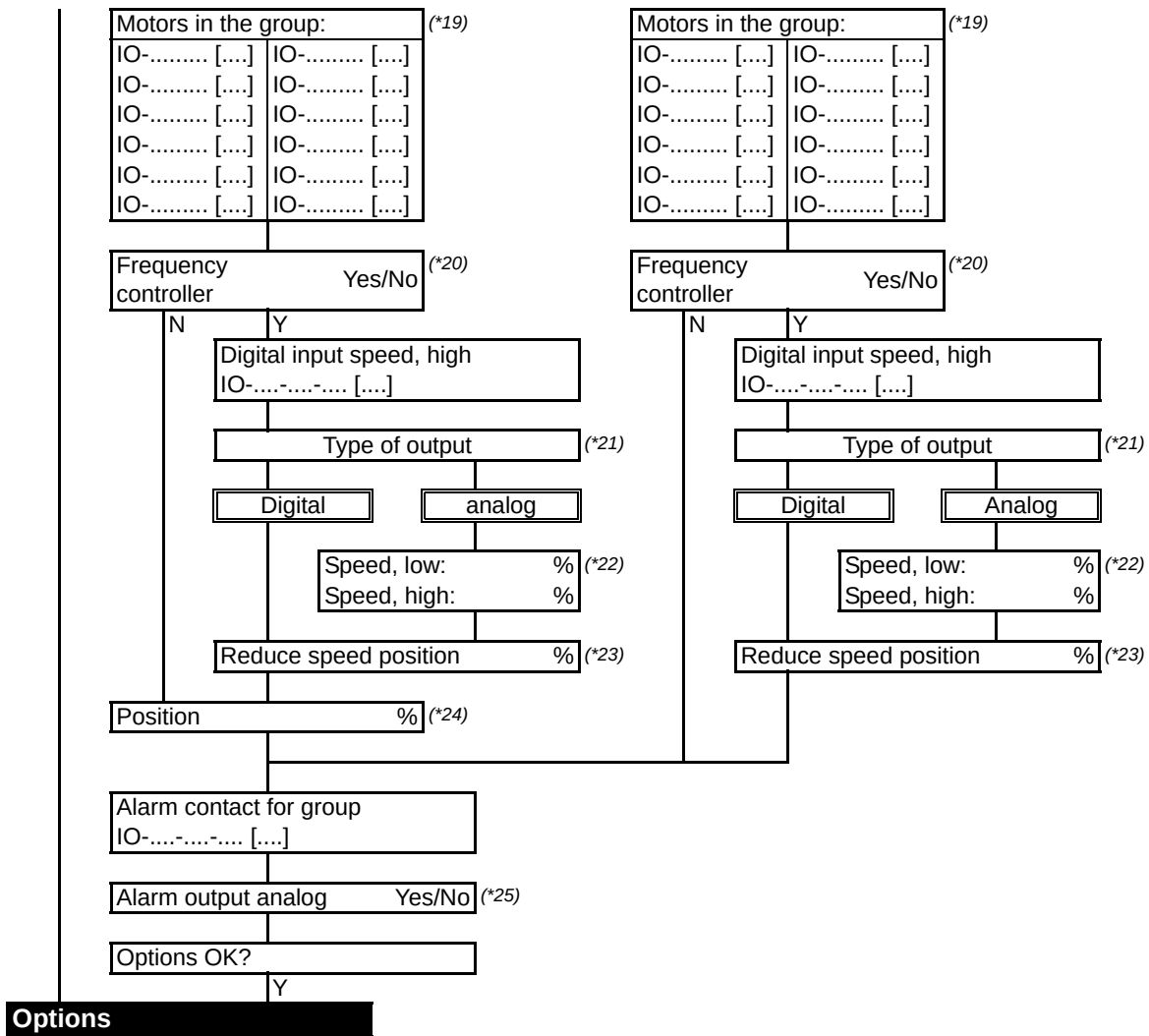
You can configure the options for each group, namely:

- window motors (airing)
- screen motors
- fan motors

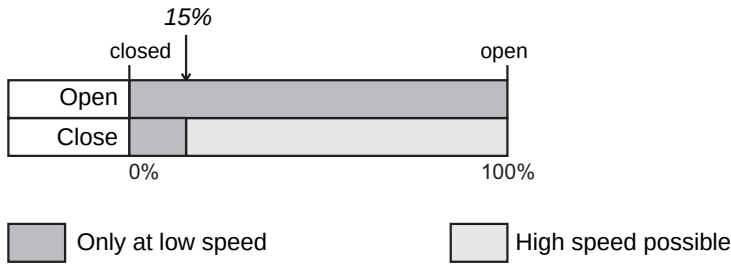


Window



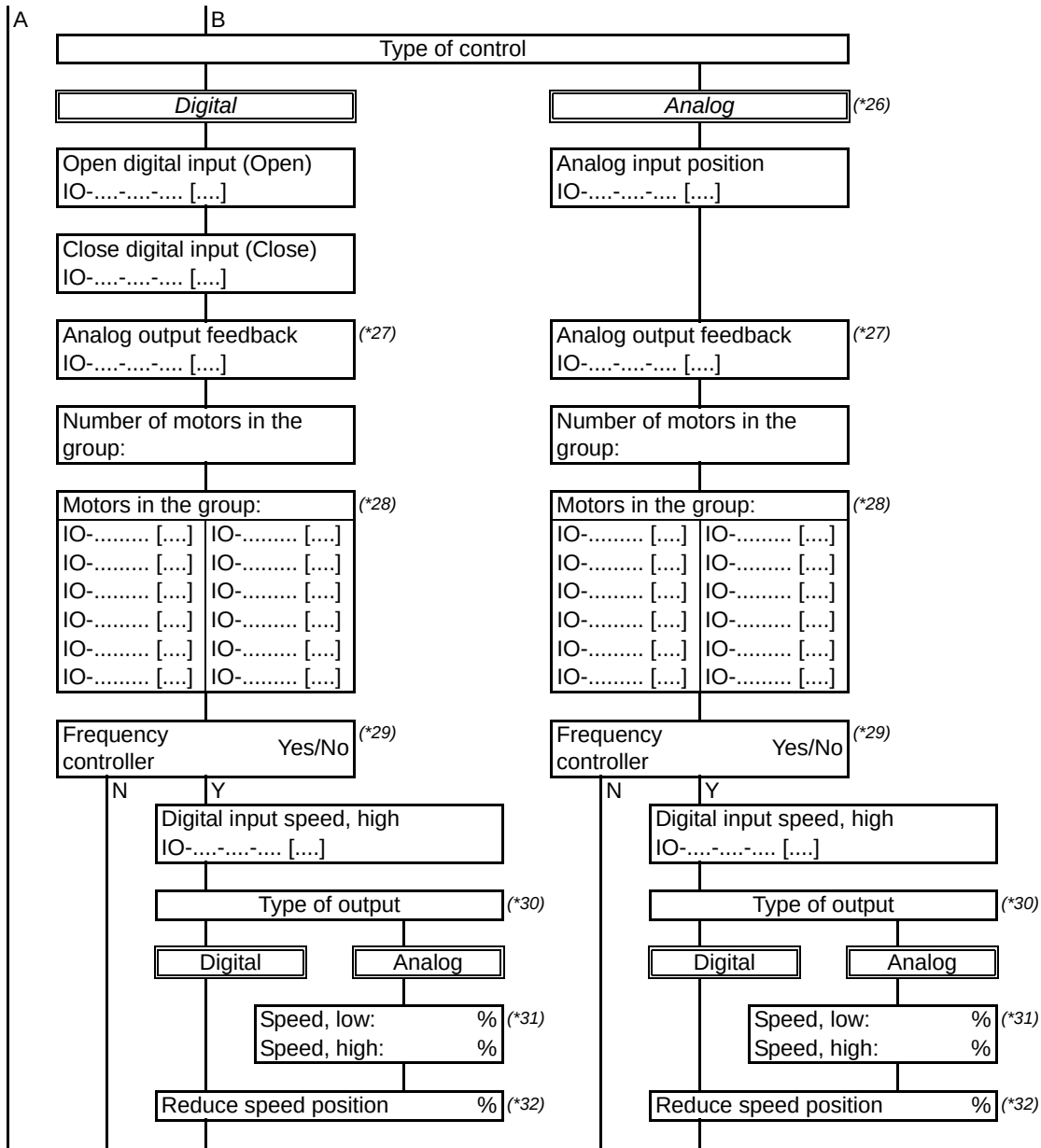


- (*17) Specify whether the control signals from the main controller are analog or digital here. These options are not applicable to CANopen or BACnet communication.
- (*18) Specify the analog output here when feedback to the main controller is necessary.
- (*19) Assign the motors belonging to this group here.
- (*20) Frequency-controlled motors can be used to close windows faster. Frequency-controlled motors can only be controlled by a H2MC-F or H1MC-PI (see also Appendix 2).
- (*21) With digital control of the frequency controller a contact on the H2MC-F or H1MC-PI is closed when the motor needs to turn at a higher speed (jog speed).
With analog control the frequency controller is controlled via the H2MC-F or H1MC-PI using a 0..5 V or 0..20 mA signal (see also (*22)).
- (*22) The signal is limited to two positions, *speed, low* and *speed, high*. **Example** when *speed, low* is 40% then the output signal at low speed is 2 V or 8 mA (40% of 5 V, or 40% of 20 mA). **The ultimate speed depends on the frequency controller settings.**
- (*23) The reduce speed position determines the position from which the window must always close at low speed. Once the measured position is equal to or less than 0% + *reduce speed position* then a high speed is not possible. **Example** *reduce speed position* = 15%:



- (*24) This option is applicable when the main controller controls the position of the window on the basis of a running time. You will then need to configure the main controller's output position. **Example** The main controller calculates that the setting of the window position should be 45.3%. Enter 45.3% under *position*.
- (*25) When *Analog alarm output* is enabled then in the event of an alarm the analog feedback signal to the main controller is set to 50%.

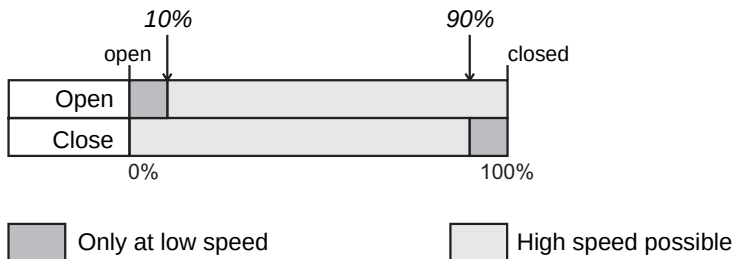
Screen



Pulse system	Yes/No	(*33)
Gap control	Yes/No	(*34)
Position	%	(*35)
Alarm contact for group IO-....-....-.... [...]		
Alarm output analog	Yes/No	(*36)
Options OK?	Y	

Options

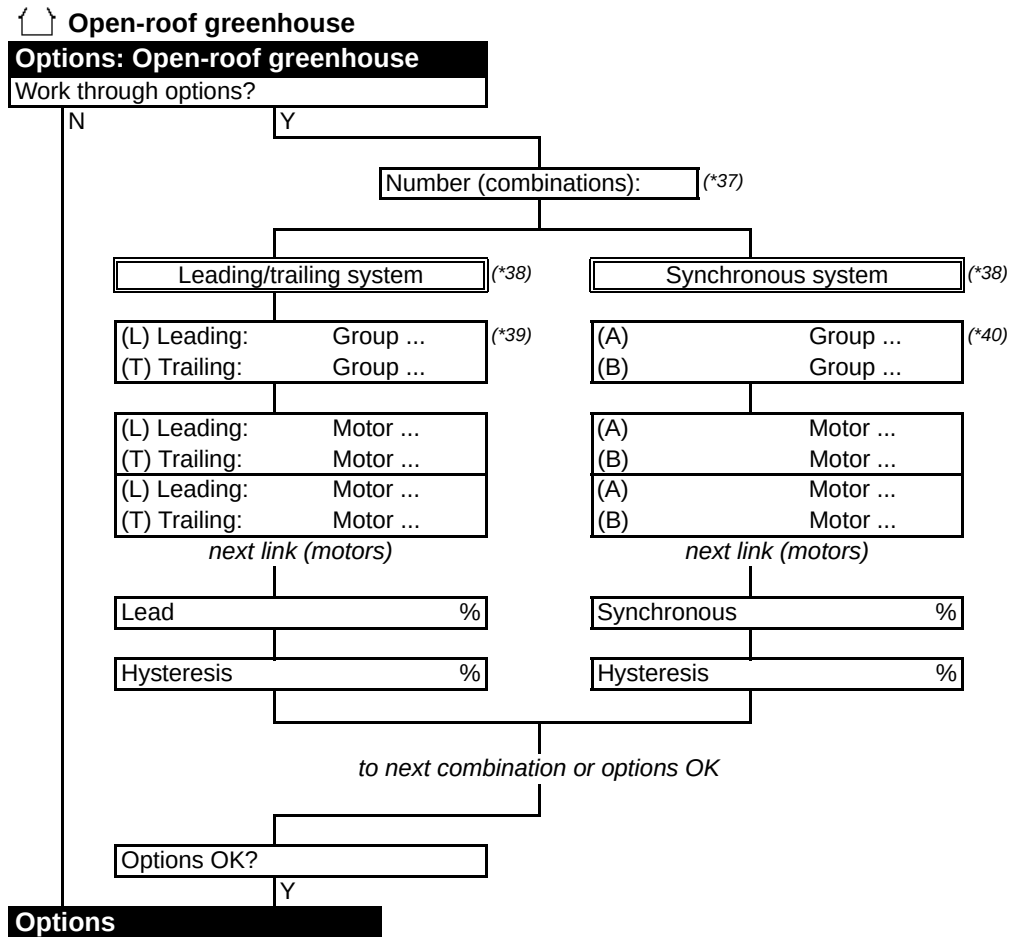
- (*26) Specify whether the control signals from the main controller are analog or digital here. These options are not applicable to CANopen or BACnet communication.
- (*27) Specify the analog output here when feedback to the main controller is necessary.
- (*28) Assign the motors belonging to this group here.
- (*29) Frequency-controlled motors can be used to close screens faster. Frequency-controlled motors can only be controlled by a H2MC-F or H1MC-PI (see also Appendix 2).
- (*30) With digital control of the frequency controller a contact on the H2MC-F or H1MC-PI is closed when the motor needs to turn at a higher speed (jog speed).
With analog control the frequency controller is controlled via the H2MC-F or H1MC-PI using a 0..5 V or 0..20 mA signal (see also (*31)).
- (*31) The signal is limited to two positions, *low speed* and *high speed*. **Example** when *speed, low* is 40% then the output signal at low speed is 2 V or 8 mA (40% of 5 V, or 40% of 20 mA). **The ultimate speed depends on the frequency controller settings.**
- (*32) The reduce speed position determines the position from which the screen must always move at low speed. While the measured position is less than 0% + *reduce speed position* or more than 100% - *reduce speed position* then a high speed is not possible. **Example** *reduce speed position* = 10%:



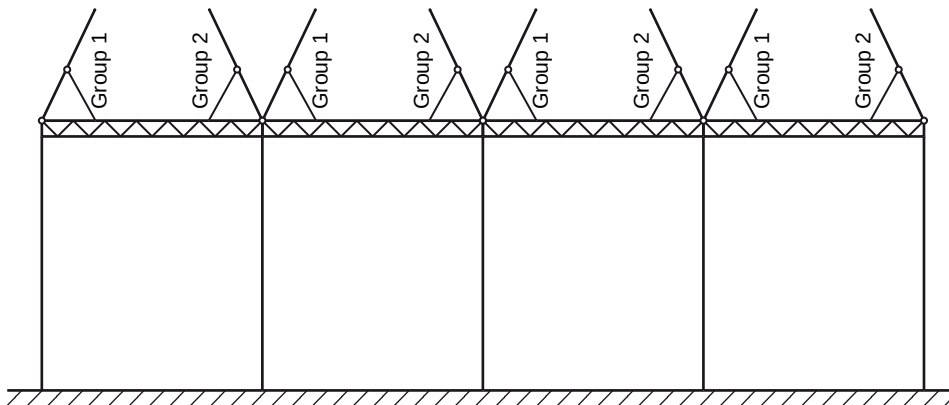
- (*33) The pulse system can be used to adjust the position of the screen to take account of the influences caused by the contraction and expansion of the greenhouse. More information is enclosed in chapter 5.1.4.
The pulse system can be used only when the screens are equipped with a slip system.
- (*34) Since the virtual motor (see also chapter 3.2) leads the actual position the main controller receives a 100% feedback before the screen has actually reached this position. The main controller can then give a command to make a gap. This makes it possible to control the screen to create a gap immediately, rather than first closing the screen (100% position). When the *Gap control* option is enabled the screen is actually sent to the 100% position before being sent to the gap position.
- (*35) This option is applicable when the main controller controls the position of the screen on the basis of a running time. You will then need to configure the main controller's output position. **Example** The main controller calculates that the setting of the screen should be 45.3%. Enter 45.3% under *position*.
- (*36) When an *analog alarm output* is enabled then in the event of an alarm the analog feedback signal to the main controller is set to 50%.

8.5 Options for open-roof greenhouse

With the *Options for open-roof greenhouse* you link motor groups of window motors to each other. More information is enclosed in Appendix 2.2.



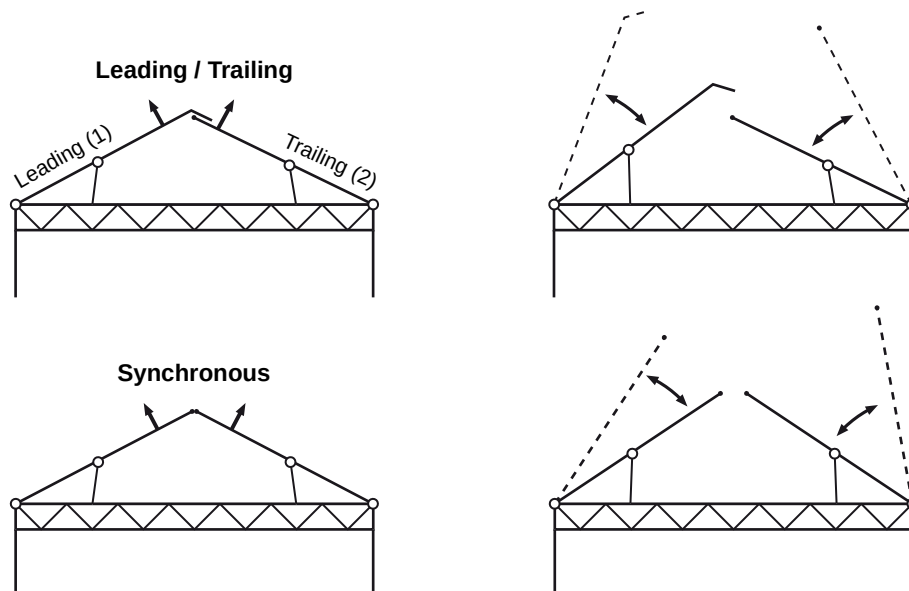
(*37) Configure the number of combinations here. A combination is comprised of two groups of window motors that belong together. The groups are for opposing windows. **Example** group 1 and group 2 form a combination in the open-roof greenhouse shown below.



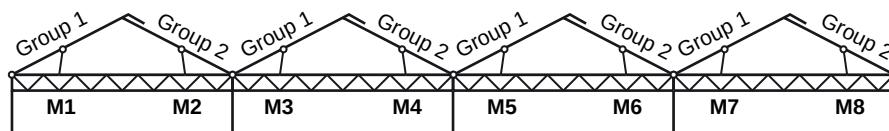
(*38) Two options are available. The choice depends on the structure of the ridge of the closed roof. *Leading / Trailing*: When the windows are opened the leading group is opened first and then the trailing group is opened. When the windows are closed the trailing group is closed first and then the leading group is closed (see also

Appendix 2.2).

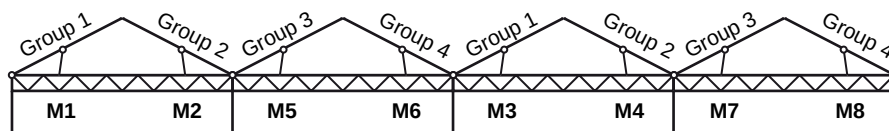
Synchronous: When the windows are opened the windows are opened synchronously for a short distance. When the windows are closed the windows are closed synchronously in the short final distance (see also Appendix 2.2).



- (*39) **Leading / Trailing.** Specify the group which is the leading group (L) and which group is the trailing group (T).
Example Group 1 = leading group (M1 + M3 + M5 + M7); group 2 = trailing group (M2 + M4 + M6 + M8). Next, specify which motors in these groups belong together. **Example** Motor M1 works together with motor M2; motor M3 works together with motor M4, etc.
 The *lead* setting determines the minimum distance that the leading group must be opened before the trailing group may open. The hysteresis serves as a safety margin to absorb possible variances. The leading group opens by a distance of at least the *lead* + *hysteresis* position.



- (*40) **Synchronous.** Specify which groups belong together. **Example** Group 1 (M1 + M3 + M5 + M7) works together with group 2 (M2 + M4 + M6 + M8). Then specify which motors in these groups belong together. **Example** Motor M1 works together with motor M2; motor M3 works together with motor M4, etc.
 The *synchronous* setting determines the section of the travel in which the groups must always move synchronously. When, for example, you enter a setting of 10% then the windows will always move synchronously in the section of the travel of between 0 and 10% both when the windows are opened and closed. The hysteresis serves as a safety margin to absorb possible variances. The group opens by a distance of at least the *synchronous* + *hysteresis* position.



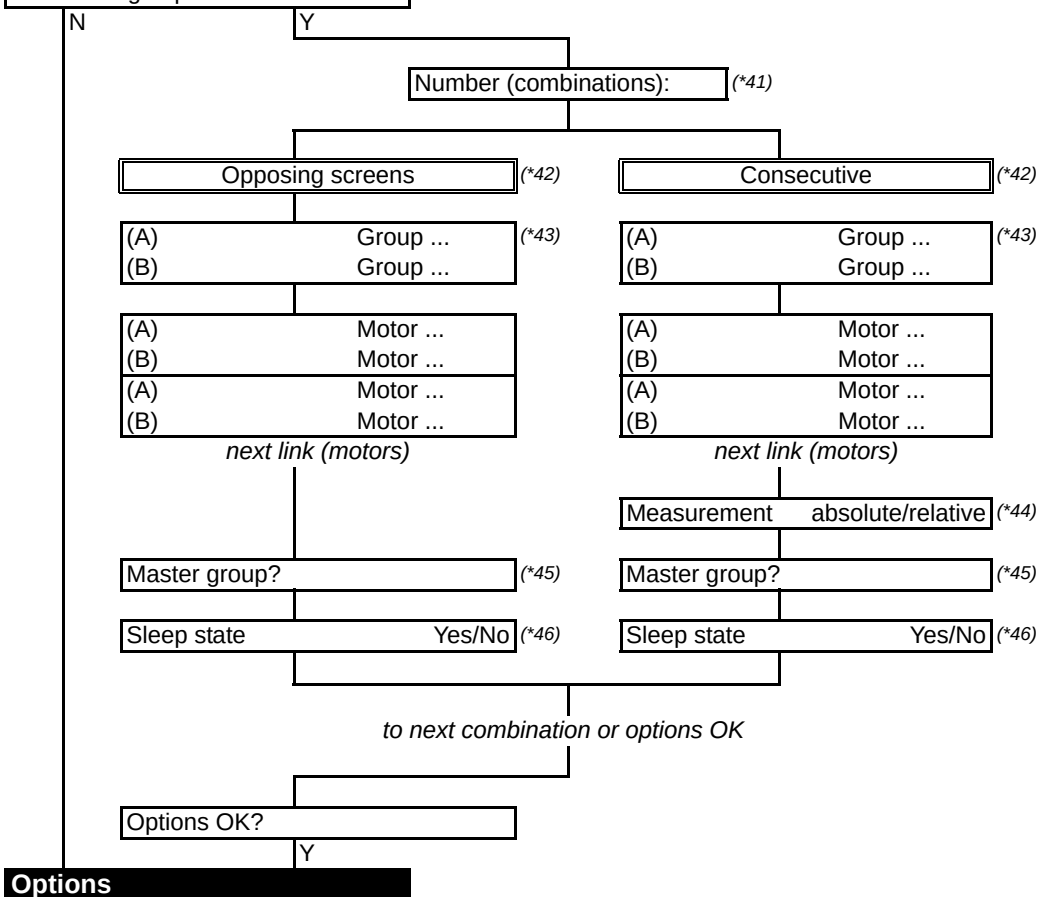
8.6 Options for 2 screens on one frame

You use the *options for 2 screens on 1 frame* to link motor groups with screen motors to each other. More information is enclosed in Appendix 2.3.

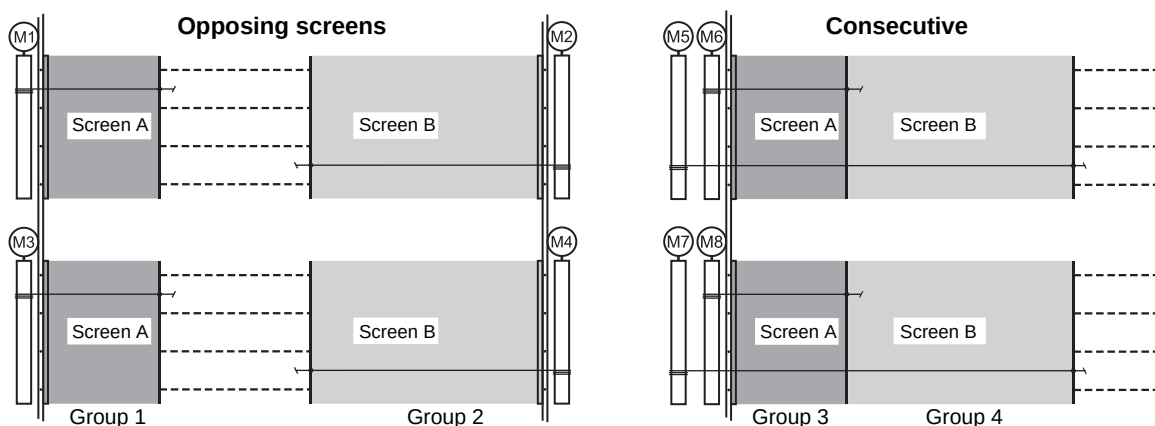
2 Screens on 1 frame

Options: 2 Screens on 1 frame

Work through options?



(*41) Configure the number of combinations here. A combination is comprised of two groups of screen motors that belong together. **Examples** Group 1 = M1 + M3; Group 2 = M2 + M4; Group 3 = M5 + M7; Group 4 = M6 + M8.



(*42) Two options are available. The choice depends on the system (see also the drawing above).

Opposing screens: The screens on the cable frame move towards each other.

Consecutive: The first screen is mounted on a joist of the greenhouse structure. The second screen is mounted on the moveable section of the first screen (see also Appendix 2.3).

(*43) Specify which groups belong together. **Example** Group 1 (M1 + M3) works together with group 2 (M2 + M4).

Note With *two consecutive screens* A is the motor that operates the front screen and B is the motor that operates the back screen (this is the screen mounted on the greenhouse joist) (see also the pictogram ).

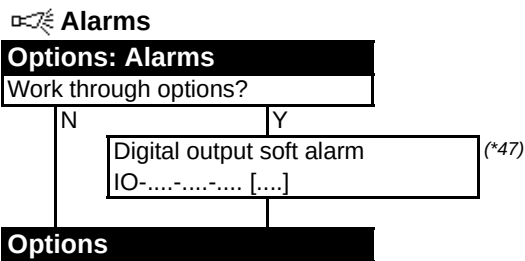
Then specify which motors in these groups belong together. **Example** Motor M1 works together with motor M2; motor M3 works together with motor M4.

(*44) Example: Screen group 1 = 40%, Screen group 2 = 30%.



- (*45) The master group always has priority over the other group when the screens are controlled simultaneously. When no master group is specified then the last motor group to be controlled has priority (see Appendix 2.3).
- (*46) When the sleep state is enabled then the gap or lead distance between the two screens is eliminated after a period of inactivity (see chapter 5.2 and chapter 5.3).

8.7 Alarms



(*47) You can assign a contact to a soft alarm here. This contact will then be de-energised in the event of a soft alarm.

9 Technical specifications

9.1 Orion-UPM-30

IMPORTANT!

- When an installation has two or more components that are fed separately, check if all components can operate at the same mains voltage.
- Always replace a defective fuse with a fuse of the same brand, type and fuse rating.

Electrical

Types of mains supply	: Fixed wiring
Operating area	: Indoor, normal, dry (not suitable for wet areas)
Operating conditions	: Continuous operation
Mains power supply	: 100-240 VAC
Frequency	: 50-60 Hz
Power consumption	: 0.75-0.5 A
PCB fuse	: T800mA (5 mm x 20 mm)
Pre-fuse	: Max. 16 A
Insulation	: Class 1 (IEC 61140)
Transient over-voltage	: Category II (IEC 60664-1)
Pollution degree	: Category 2 (IEC 61010-1)

Ethernet communication (between Orion-CPU and main controller)

Protocol	: BACnet
Cable required:	: UTP cat 5e
Maximum length:	: 100 metres @ 10 Mbps

Mechanical

Mobility	: Fixed position (the product is mounted to the wall)
Operating temperature range	: 0 - 40 °C
Relative humidity	: Max. 80%
Installation height	: Max. 2000 m above sea level
Dimensions (H x W x D)	: 220 x 165 x 105 mm
Housing	: IP54, ABS
Weight	: ± 1 kg

Power supply (UPM-30)

Voltage 24 VDC $\pm 2.5\%$	: Max. 800 mA
----------------------------	---------------

Communication

CAN Local	: Max. 500 m @ 100 kbps
CAN Backbone	: Max. 500 m @ 125 kbps
RS232	: Max. 15 m @ 38400 baud
USB	: Max. 5 m @ 115 kbps
Ethernet	: Max. 100 m @ 10 Mbps

Cables

CAN Local / CAN Backbone	: Shielded twisted pair, 1x2x0.64 mm ² (CAN-BK cable)
RS232	: MSK-9
USB	: USB with USB-B mini plug and USB-A plug
Ethernet	: UTP cat 5e

Compliance

EU Directives	: See the Declaration of Conformity for this product
---------------	--

9.2 CAN-IO-08-09P

IMPORTANT!

- This product can have more than one power source. Always disconnect all the power sources before you do work on live or exposed parts of the product.
- When an installation has two or more components that are fed separately, check if all components can operate at the same mains voltage.
- Always replace a defective fuse with a fuse of the same brand, type and fuse rating.
- Always prevent ground loops when you install the product. If possible, provide galvanic isolation.

Electrical

Types of mains supply	: Fixed wiring
Operating area	: Indoor, normal, dry (not suitable for wet areas)
Operating conditions	: Continuous operation
Mains power supply	: 230 VAC $\pm 10\%$
Frequency	: 50/60 Hz
Power consumption	: max. 15 VA
PCB fuse	: T100mA (5 mm x 20 mm)
Insulation	: Class 1 (IEC 61140)
Transient over-voltage	: Category II (IEC 60664-1)
Pollution degree	: Category 2 (IEC 61010-1)

Mechanical

Mobility	: Fixed position (the product is mounted to the wall)
Operating temperature range	: 0 - 40 °C
Relative humidity	: Max. 80%
Installation height	: Max. 2000 m above sea level
Dimensions (H x W x D)	: 220 x 165 x 105 mm
Housing	: IP54, ABS
Weight	: ± 1 kg

Ethernet communication (between Orion-CPU and main controller)

Protocol	: BACnet
Cable required:	: UTP cat 5e
Maximum length:	: 100 metres @ 10 Mbps

Digital inputs (8)

Contact / counter	: NPN / PNP sensor, 12-24 VDC, 8 mA, max. 10 Hz
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Analog outputs (4)

0-5 VDC	: Max. 1 mA
0-20 mA	: Max. 10 VDC

Digital outputs (4 + 1 alarm output)

Relay output K1-K4 : 0.5 A, 230 VAC
Alarm output : 0.5 A, 24 VAC/DC

Power supply, 24 VDC

Voltage 24 VDC -30% +10%t : Max. 200 mA

Communication

CAN Local : Max. 500 m @ 100 kbps

Cables

CAN Local : Shielded twisted pair, 1x2x0.64 mm² (CAN-BK cable)

Compliance

EU Directives : See the Declaration of Conformity for this product

10 Connection diagrams

10.1 Connection sequence

Read these instructions thoroughly before you begin work on the assembly and connections.

This product should be installed by an approved installer with certification according to the relevant prevailing standards (for example, ANSI, BS, ISO). The installer must possess the knowledge of and experience with electronic control equipment required to install the Orion-MC.

IMPORTANT! For your safety, you must work through the installation procedure in the following sequence.

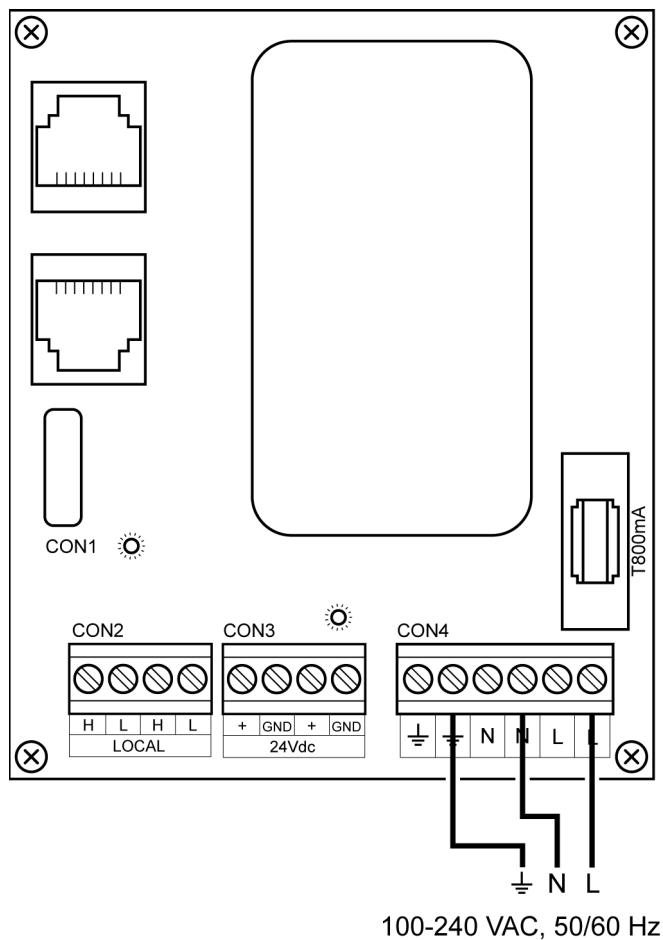
1. Install the necessary equipment (see also Appendix 2).
2. Install the necessary IO modules (see also Appendix 2).
3. Read in all motors and equipment with position feedback: these are the motors controlled using the IO modules such as the H1MC, H2MC and CAN-IO-EKU modules. Reading in these motors and equipment will ensure that they do not go out of range during the installation of the Orion-MC.
4. If possible, set all IO modules and connected equipment in manual operation mode: this will make sure that the motors and equipment to be controlled by the unit will not be started unintentionally during the installation of the Orion-MC.

Note The following connection diagrams explain the communication with the main controller, and do not extend to the connection of options (such as temperature sensors, RS485 communication, etc.). Information about the connection of options is enclosed in the relevant CAN IO module manual. In addition, see the examples of connections in Appendix 2.

10.2 Orion-UPM-30

IMPORTANT!

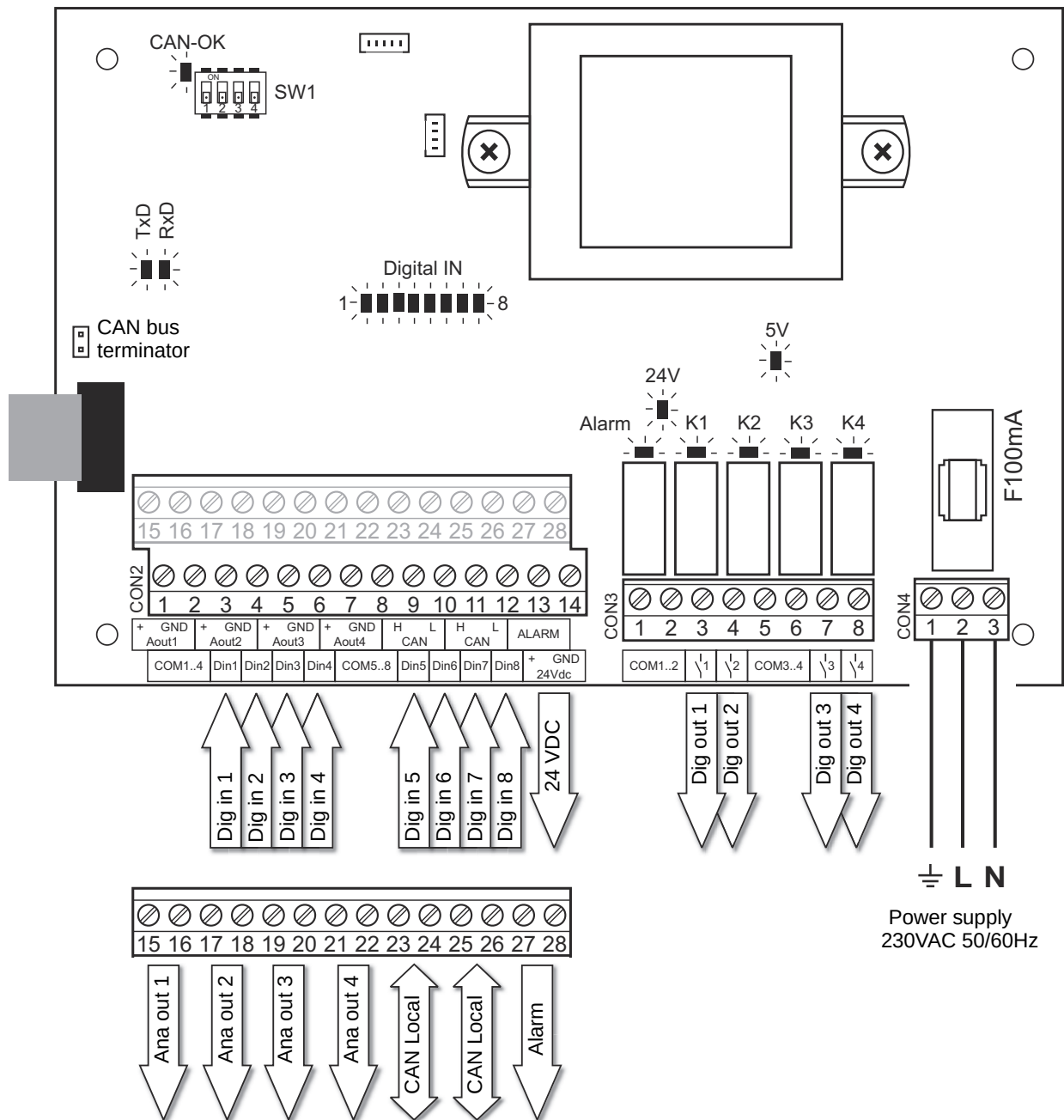
- Always disconnect the power supply to the product before you do work on electrical parts inside the product. When this is not possible, be very careful that you do not touch live parts.
- This product can have more than one power source. Always disconnect all the power sources before you do work on live or exposed parts of the product.
- When an installation has two or more components that are fed separately, do a check if all the components can operate at the same mains voltage.
- All the electrical parts inside this product are susceptible to electrostatic discharge (ESD).
 - Always disconnect the power supply to the product before you do work on electrical parts inside the product.
 - Always wear an ESD wrist strap that is connected to the main earthing system. See IEC61340-5-2 for more information on precautions with regard to ESD.
- Always prevent ground loops when you install the product. If possible, provide galvanic isolation.
- A mix of voltages at the relay contacts can damage your product.
 - Connect the digital outputs (relay contacts) to the same electrical potential.
 - Do not mix mains and low-voltage signals.



10.3 CAN-IO-08-09P

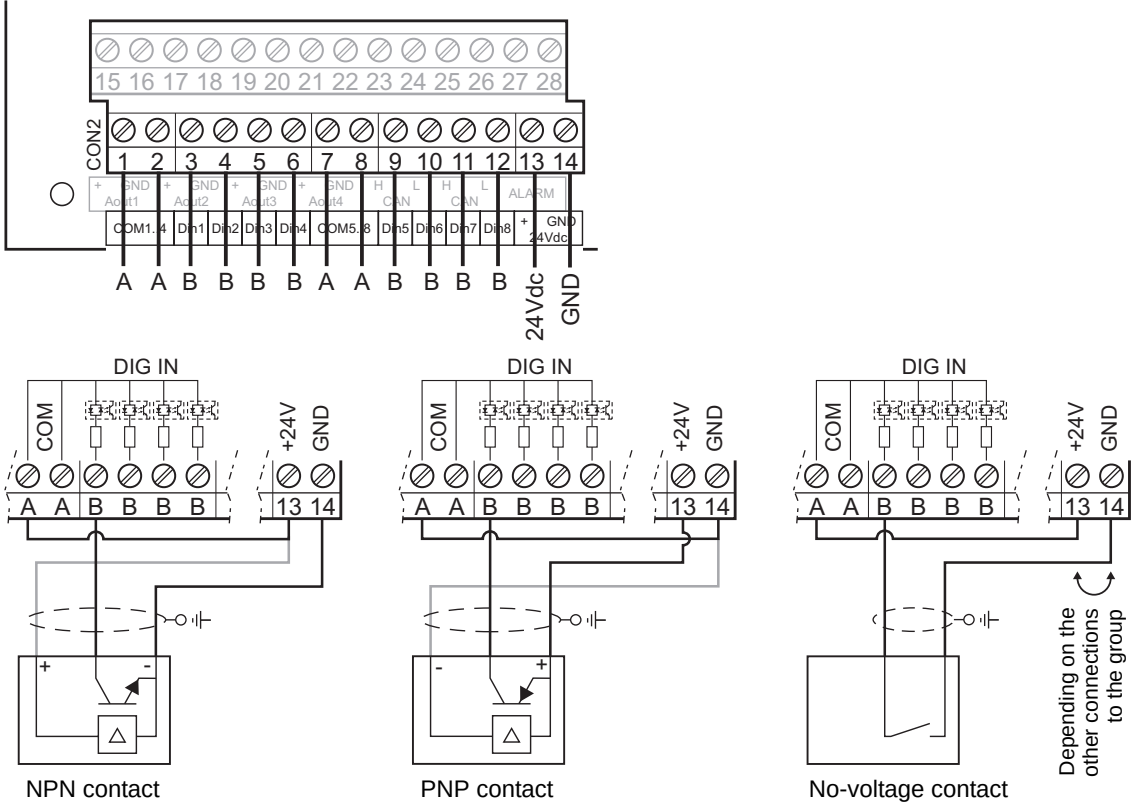
IMPORTANT!

- Always disconnect the power supply to the product before you do work on electrical parts inside the product. When this is not possible, be very careful that you do not touch live parts.
- This product can have more than one power source. Always disconnect all the power sources before you do work on live or exposed parts of the product.
- When an installation has two or more components that are fed separately, do a check if all the components can operate at the same mains voltage.
- All the electrical parts inside this product are susceptible to electrostatic discharge (ESD).
 - Always disconnect the power supply to the product before you do work on electrical parts inside the product.
 - Always wear an ESD wrist strap that is connected to the main earthing system. See IEC61340-5-2 for more information on precautions with regard to ESD.
- Always prevent ground loops when you install the product. If possible, provide galvanic isolation.
- A mix of voltages at the relay contacts can damage your product.
 - Connect the digital outputs (relay contacts) to the same electrical potential.
 - Do not mix mains and low-voltage signals.



10.3.1 Digital inputs

The digital inputs are intended for the control signals from the main controller. See Appendix 2.1 for examples of connections. The digital inputs are assigned during the installation procedure.



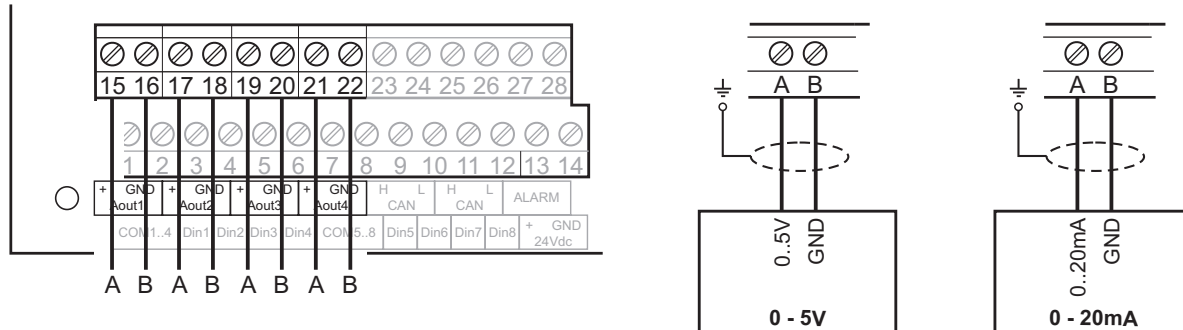
Note You can connect either NPN contacts or PNP contacts to one input group. Example: When a PNP contact is connected to digital input 6 then you may only connect other PNP contacts or no-voltage contacts to inputs 5, 7 and 8.

Note the inputs in use in the following table.

Digital input	Terminal			Selection								Description
				Window	Screen	Ventilation	Flap	Heating	Alarm			
1	1/2	-	3									
2	1/2	-	4									
3	1/2	-	5									
4	1/2	-	6									
5	7/8	-	9									
6	7/8	-	10									
7	7/8	-	11									
8	7/8	-	12									

10.3.2 Analog outputs

The analog outputs are intended for the feedback to the main controller. See Appendix 2.1 for examples of connections. The analog outputs are assigned during the installation procedure.



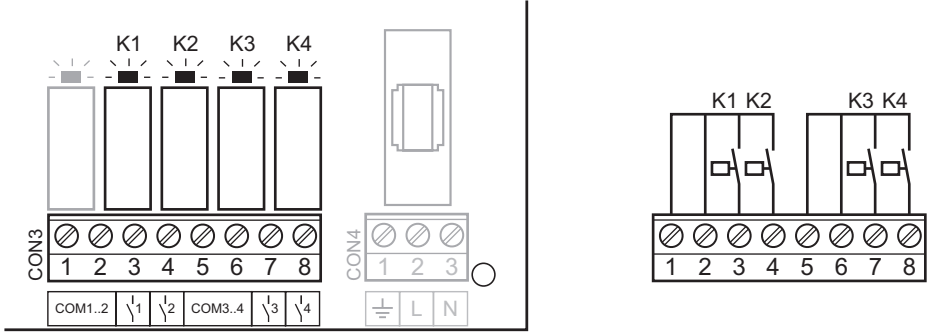
Note the outputs in use in the following table. Also note the type of output that is used (0-5V or 0-20mA) and the range of the output (for example, 4-20mA).

Analog output	Terminal A - B	Selection						Type of output	Range Min / Max	Description
		Window position	Screen position	Fan speed	Flap position	Heater intensity				
1	15 - 16							0-5V / 0-20mA		
2	17 - 18							0-5V / 0-20mA		
3	19 - 20							0-5V / 0-20mA		
4	21 - 22							0-5V / 0-20mA		

Note Configure the type of output and the range during the installation procedure, see chapter 8.1.

10.3.3 Digital outputs

The digital outputs are intended for the feedback to the main controller. See Appendix 2.1 for examples of connections. The digital outputs are assigned during the installation procedure.

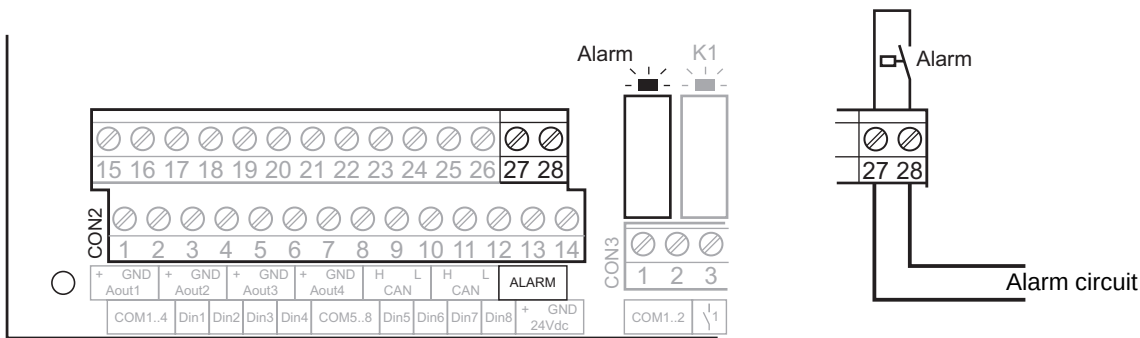


Note the outputs in use in the following table.

Digital output	Terminal	Selection								Description
		Alarm	Ventilation	Flap	Heating					
1	1/2 - 3									
2	1/2 - 4									
3	5/6 - 7									
4	5/6 - 8									

10.3.4 Alarm contact

The no-voltage alarm contact is interrupted in the event of an alarm or a power failure. You can include the alarm contact in an alarm circuit.

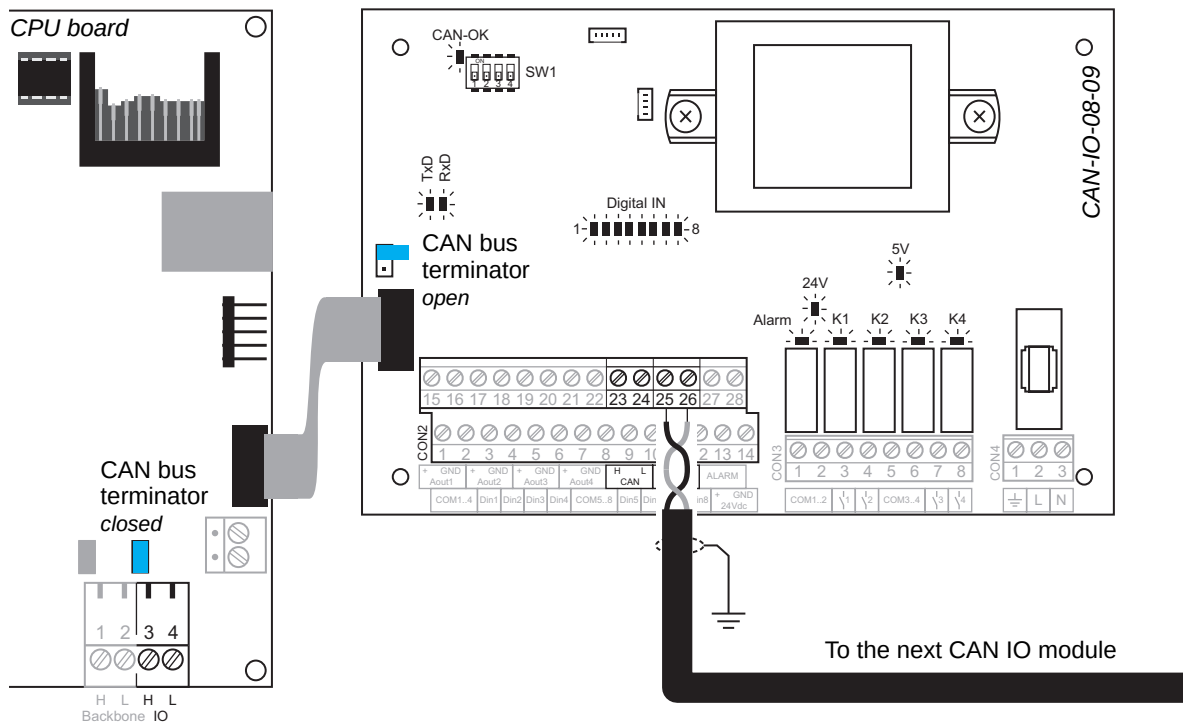


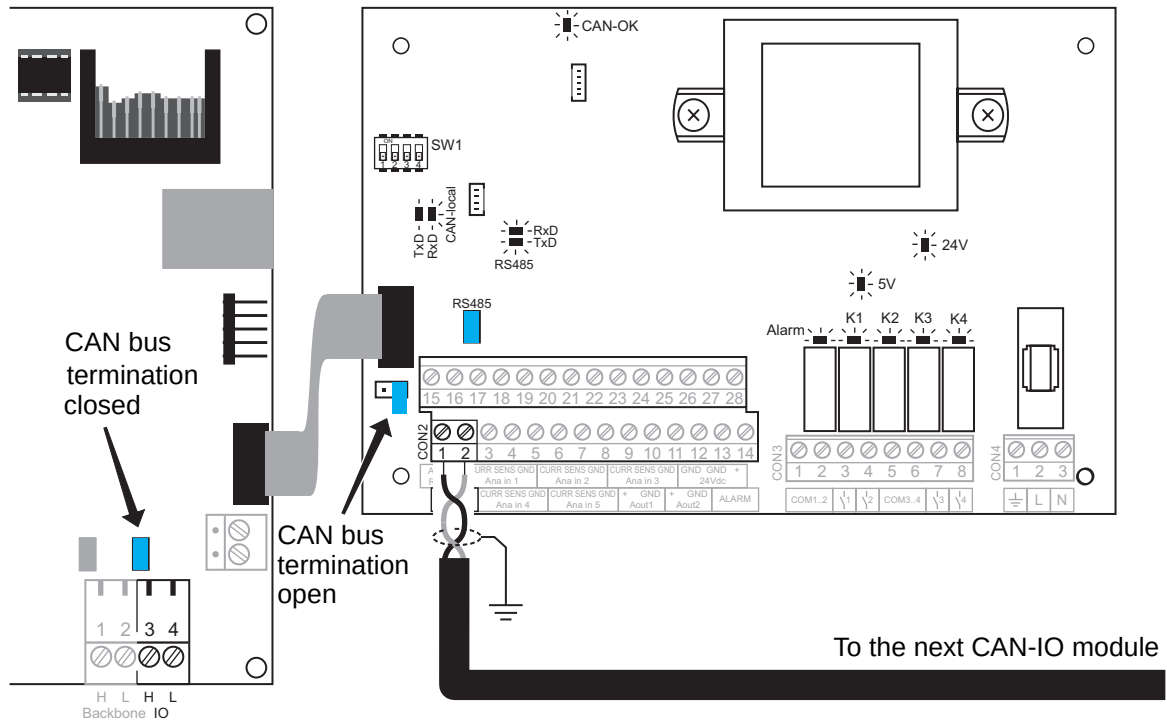
Note Configure the range during the installation procedure, see chapter 8.1.

10.4 Communication

10.4.1 CAN Local (IO modules)

The CAN Local data bus is used for communication between the Orion-MC and the IO modules (see also the examples of connections in Appendix 2).

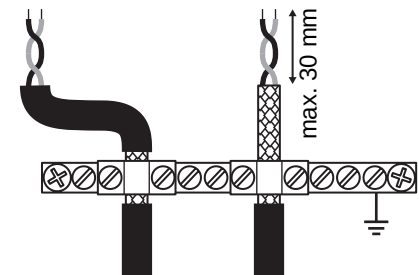




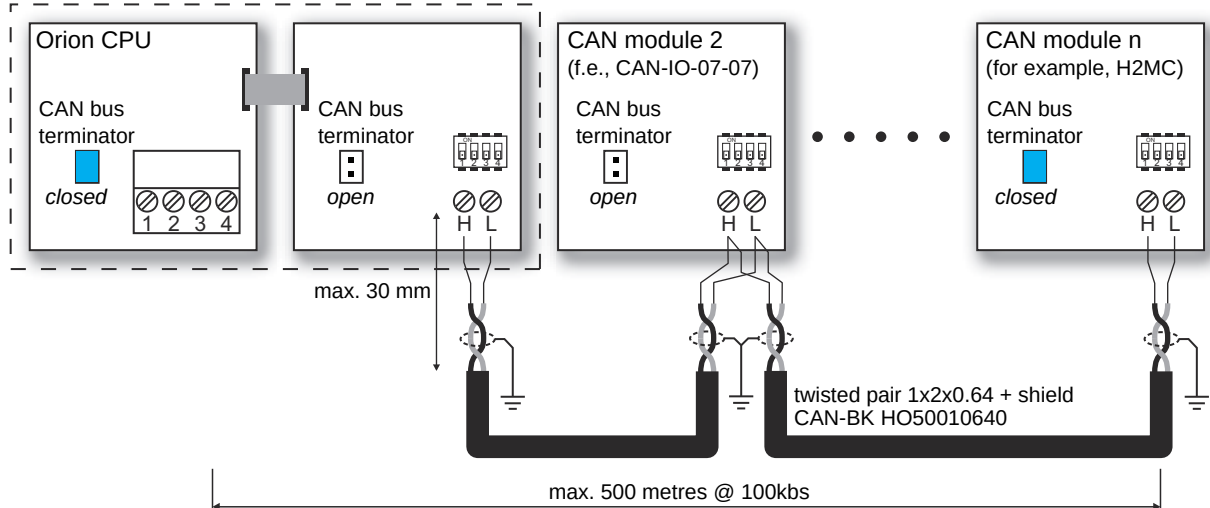
The bus must be terminated with jumpers at the first and last communication node.

Example Fit the jumpers on the CPU board and on the last IO module in the event that the CAN IO modules are connected to the integral IO module of the Orion-MC (as shown in the example above).
(Note: remove the jumper from the integral IO module or relocate it to a position where the pins are not connected.)

Example of the shield connection



Orion-MC



Note The Orion-MC does not need to be installed at the end of the communication line.

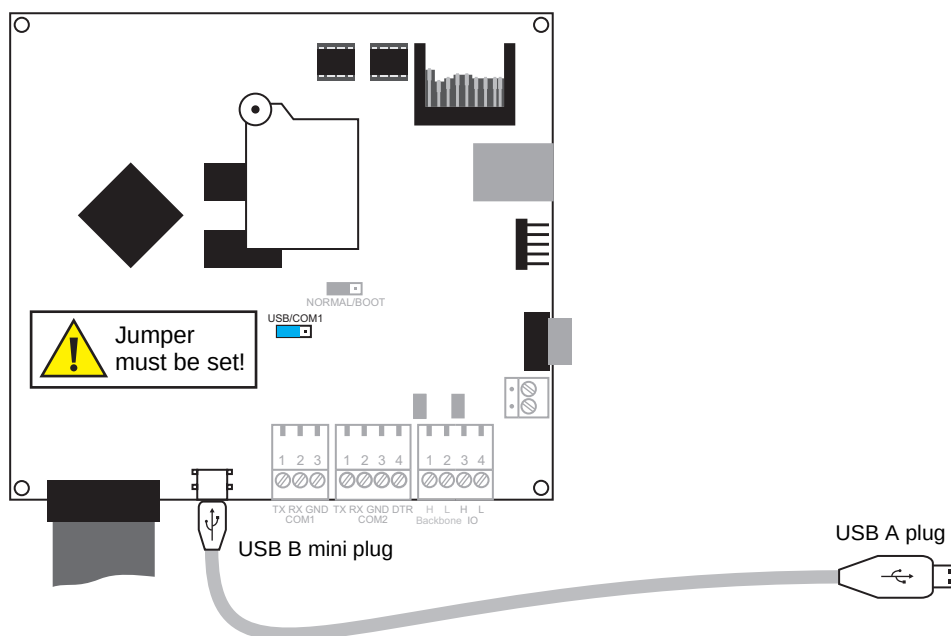
Each IO module in the communication line should be assigned a unique address. The address is configured using the IO module's DIP switch.

Address	DIP switch	Address	DIP switch	Address	DIP switch	Address	DIP switch
1		5		9		13	
2		6		10		14	
3		7		11		15	
4		8		12		16	

10.4.2 Communication with PC or Modem via USB

You can use the COM1 port or USB port (see also chapter 10.4.3). You cannot use the two ports together.

Note It is not possible to access a number of Orion-MC computers at the same time with USB.



IMPORTANT! Set the USB/COM1 jumper to USB.

Two drivers will need to be installed for communications between the Orion-MC and a PC via USB. The general instructions for a PC with Windows are given below. Begin by installing the 'USB serial converter' driver followed by the 'USB serial port' driver.

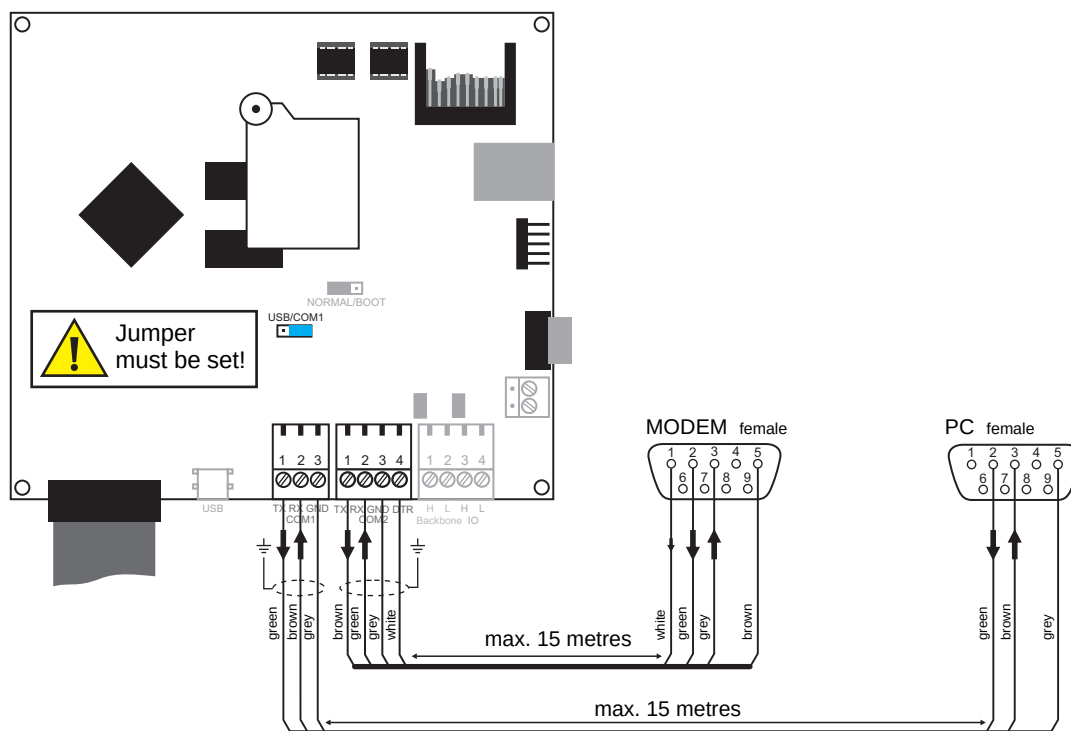
1. Connect the Orion-MC to the PC with an USB cable.
2. Connect the Orion-MC to the power supply.
3. If Windows does not recognise the USB chip then the 'New hardware found' Wizard will start.
4. Use the Wizard to search for the drivers in the CD's 'USB' folder.

5. Install the 'USB serial converter' driver (this is selected automatically). Ignore any Windows message stating that the driver is unsigned or untested.
6. Windows will now once again find new hardware.
7. Install the 'USB serial port' driver using the same method.

10.4.3 Communication with PC or Modem via RS232

You can use the COM1 port or USB port (see also chapter 10.4.2). You cannot use the two ports together. The COM2 port can be used at any time.

Note It is not possible to access a number of Orion-MC computers at the same time with RS232.

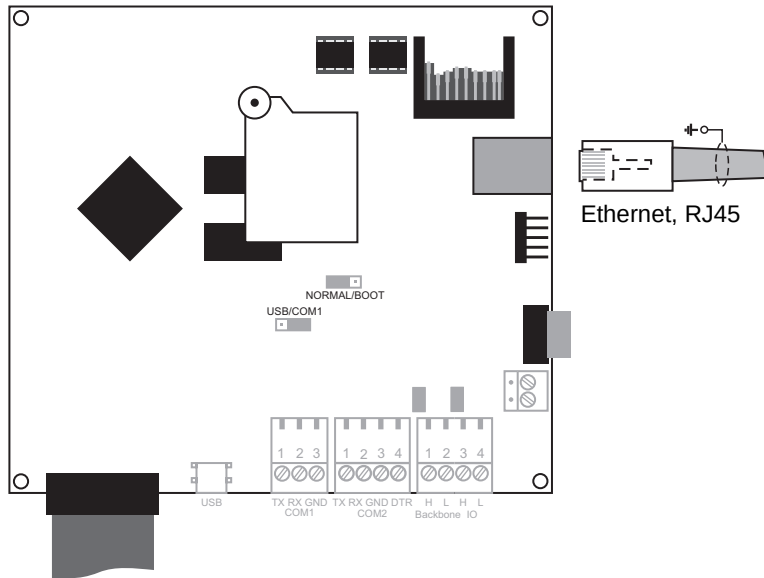


IMPORTANT! Set the USB/COM1 jumper to COM1.

Note See chapter 10.4.1 for an example of the connection of the shield.

10.4.4 Communication with PC or Modem via Ethernet

You can use Ethernet to include the Orion-MC in a local IP network.



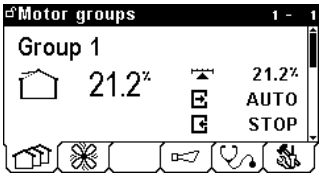
10.4.5 BACnet

BACnet can be used for communication between the main controller (horticulture computer) and the Orion-MC (see also the examples of connections in Appendix 2). BACnet communications are carried out using the Ethernet, see chapter 10.4.4.

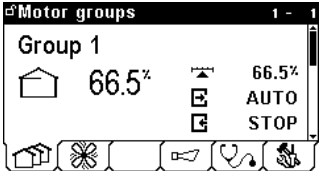
Note BACnet can be used only when the BACnet module has been activated.

Appendix 1 Settings lists

Appendix 1.1 Windows

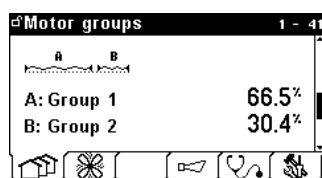
Main group	Settings	Default	User
 			
	 Operation  Position  Motor 1  Motor 2  Motor 3  Motor ...	AUTO 0.0 % 0.0 % 0.0 % 0.0 % 0.0 %	 % % % % %
 Motor ...	 Status  Operation  Position	- AUTO 0.0 %	 %

Appendix 1.2 Screens

Main group	Settings	Default	User
 			
	 Operation  Position  Motor 1  Motor 2  Motor 3  Motor ...	AUTO 0.0 % 0 % 0 % 0 % 0 %	 % % % %
 Motor ...	 Status  Operation  Position	- AUTO 0.0 %	 %

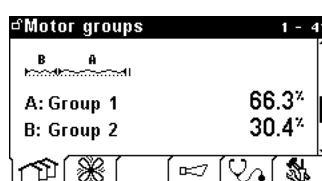
Main group	Settings	Default	User
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Two opposing screens on one cable frame



Gap	1.0 %	%
Hysteresis	0.5 %	%
Standby	30 min	min

Two consecutive screens on one cable frame

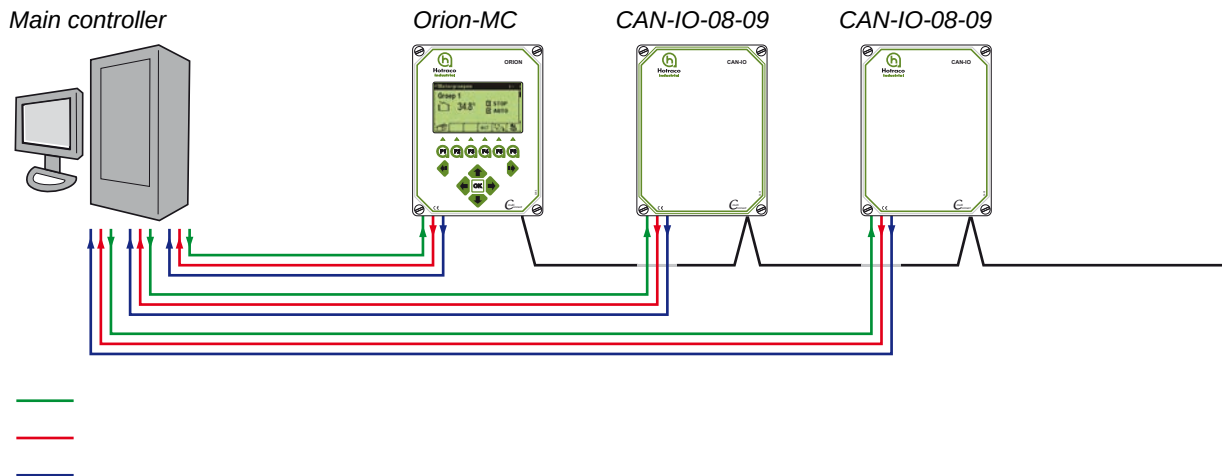


Lead	1.0 %	%
Hysteresis	0.5 %	%
Standby	30 min	min

Appendix 2 System examples

Appendix 2.1 Connection to the main controller

Digital control (Orion-MC with CAN-IO-08-09)

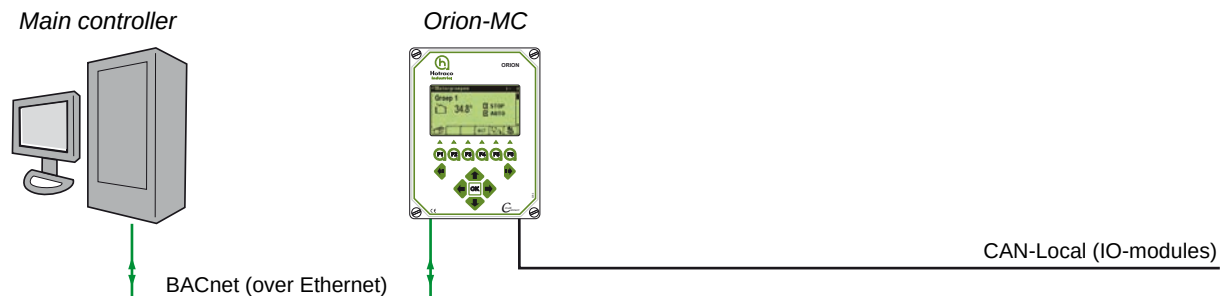


The main controller sends digital commands for each group to the Orion-MC (open, close, increase, decrease, etc.). The virtual motor of the Orion-MC sends the set position as an analog signal back to the main controller. The main controller stops sending the digital command once the virtual motor has adopted the required position. Digital feedback is given for alarms (the contact is de-energised in the event of an alarm).

Note

- You need two digital inputs and one analog output for each group: you need one digital output for each group for alarm messages.
- You can expand the number of inputs and outputs by installing extra CAN-IO-08-09 modules.

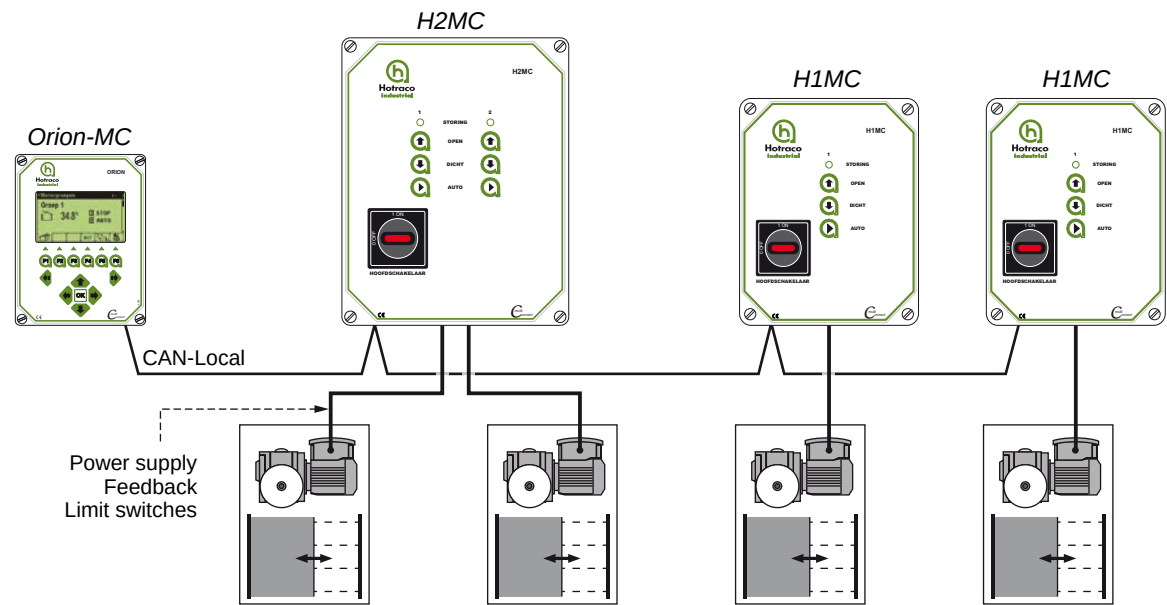
BACnet



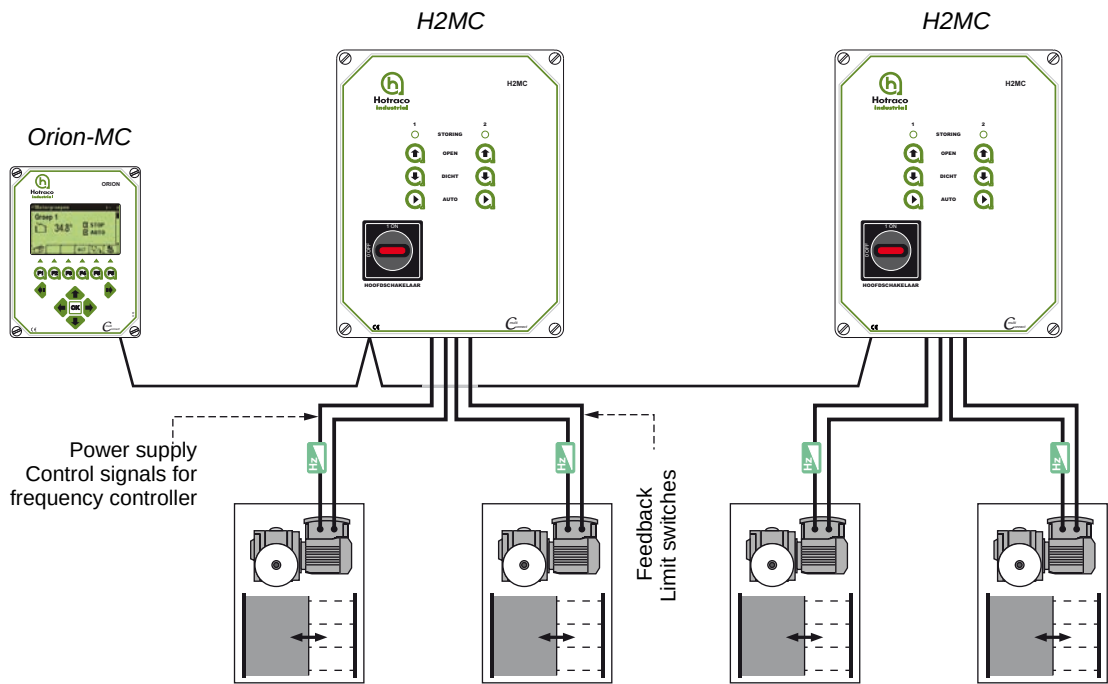
With a BACnet connection, the control signals and feedback signals are sent via an Ethernet link.

Appendix 2.2 Airing

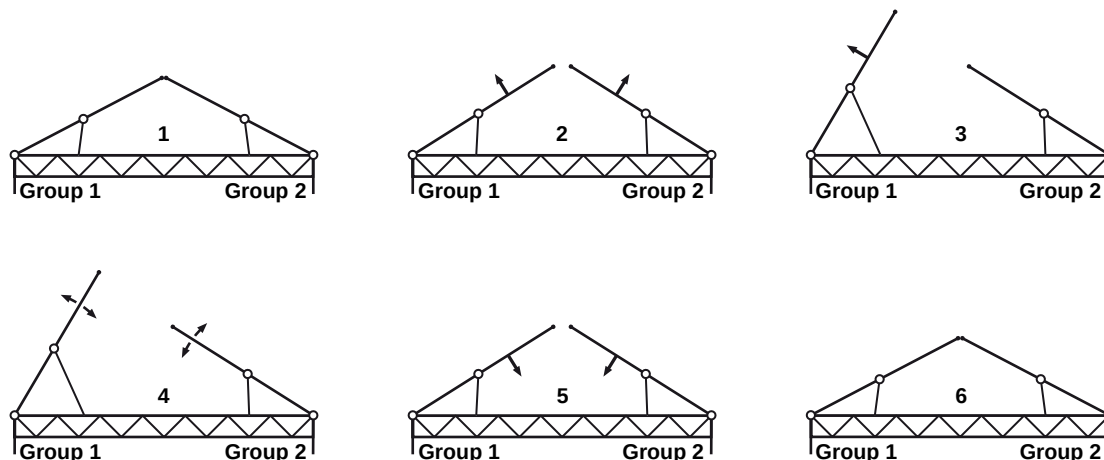
IO-modules H2MC and H1MC (example)



IO-module H2MC with frequency controller (example)



Open-roof greenhouse, synchronous (example)

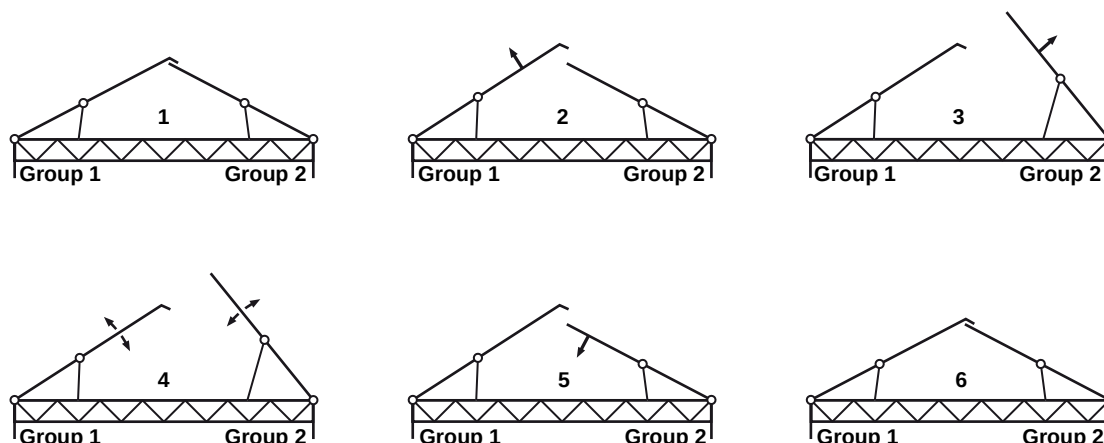


1. Group 1 and group 2 are fully closed (the position of the windows is 0%). The main controller sends a command to open group 1 to a given position.
2. Group 1 and group 2 initially move synchronously. Group 2 stops when it has reached the *synchronous position + hysteresis* (configured during the installation procedure).
3. Group 1 continues to the required position.
4. As long as group 1 and group 2 both are in a position above the *synchronous position + hysteresis*, both groups can move independently. However, the minimum position is restricted to *synchronous position + hysteresis*.
5. The main controller sends a command to close both groups completely. Group 1 and group 2 are then first sent to the *synchronous position + hysteresis*.
6. Only when both groups have reached this position, the windows will be closed simultaneously.

Note The Orion-MC makes sure that the total desired opening is maintained.

Example: Desired position group 1 = 5%, desired position group 2 = 80% and the *synchronous position + hysteresis* = 10%. Group 1 is then set to the position 10%, and group 2 to the position 75%. The total opening is then 85% (= 5% + 80%).

Open-roof greenhouse, leading, trailing



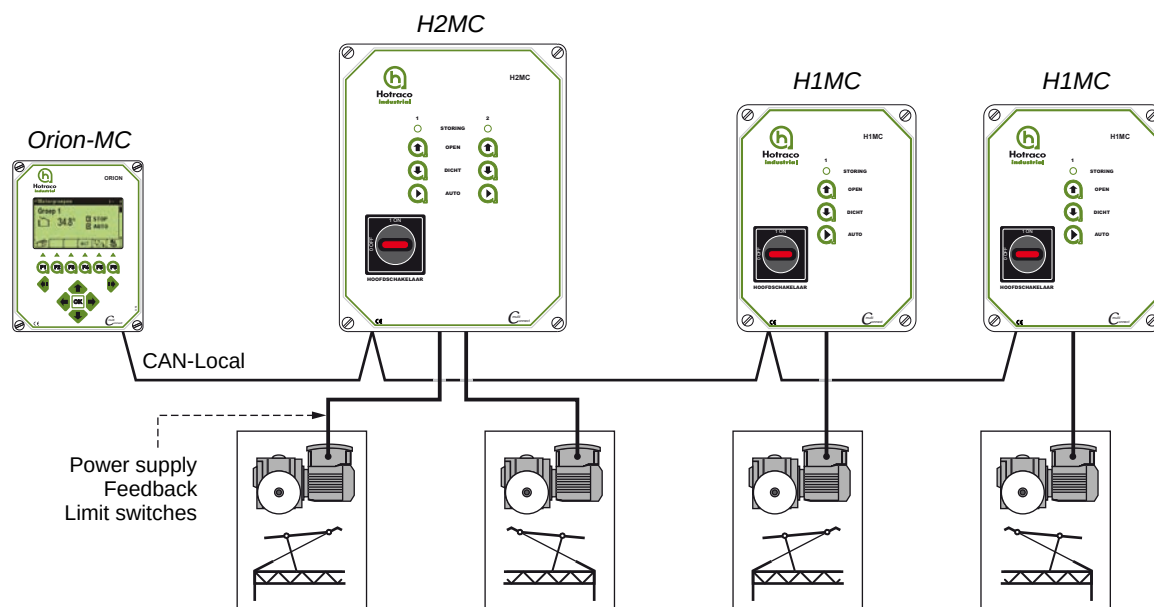
1. Group 1 is the leading group, and group 2 is the trailing group (configured during the installation procedure).
Group 1 and group 2 are fully closed (the position of the windows is 0%). The main controller sends a command to open group 2 to a given position.
2. To open group 2, group 1 will open first until it is in the *leading position + hysteresis* (configured during the installation procedure).
3. Then group 2 is set to the desired position.
4. As long as group 1 is in a position above the *leading position + hysteresis*, both groups can move independently. However, the minimum position of group 1 is restricted to the *leading position + hysteresis*.
5. The main controller sends a command to close both groups completely. Both groups get a command to close. Group 1 stops in the *leading position + hysteresis*. Group 2 is closed completely.
6. Group 1 is closed after group 2 is closed completely, and the limit switch is activated.

Note The Orion-MC makes sure that the total required opening is maintained.

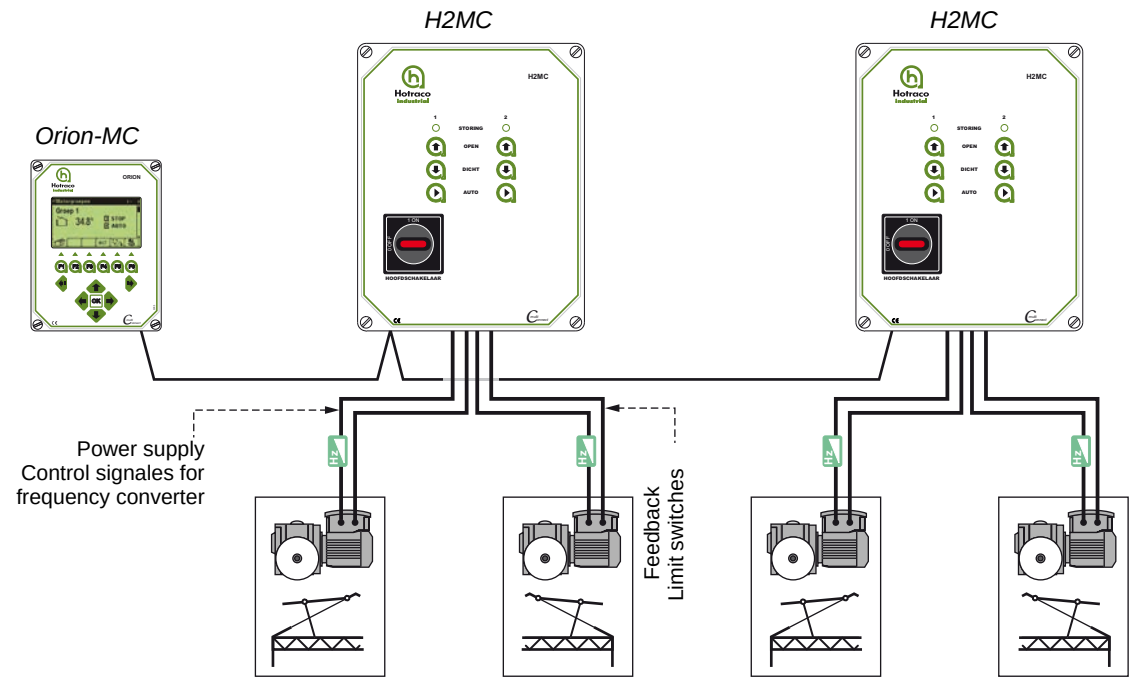
Example: Desired position of leading group = 5%, desired position of the trailing group = 80%, and the *leading position + hysteresis* = 10%. In this case, the leading group is in the position 10%. The trailing group is in the position 75%. Then the total opening is 85% (= 5% + 80%).

Appendix 2.3 Screens

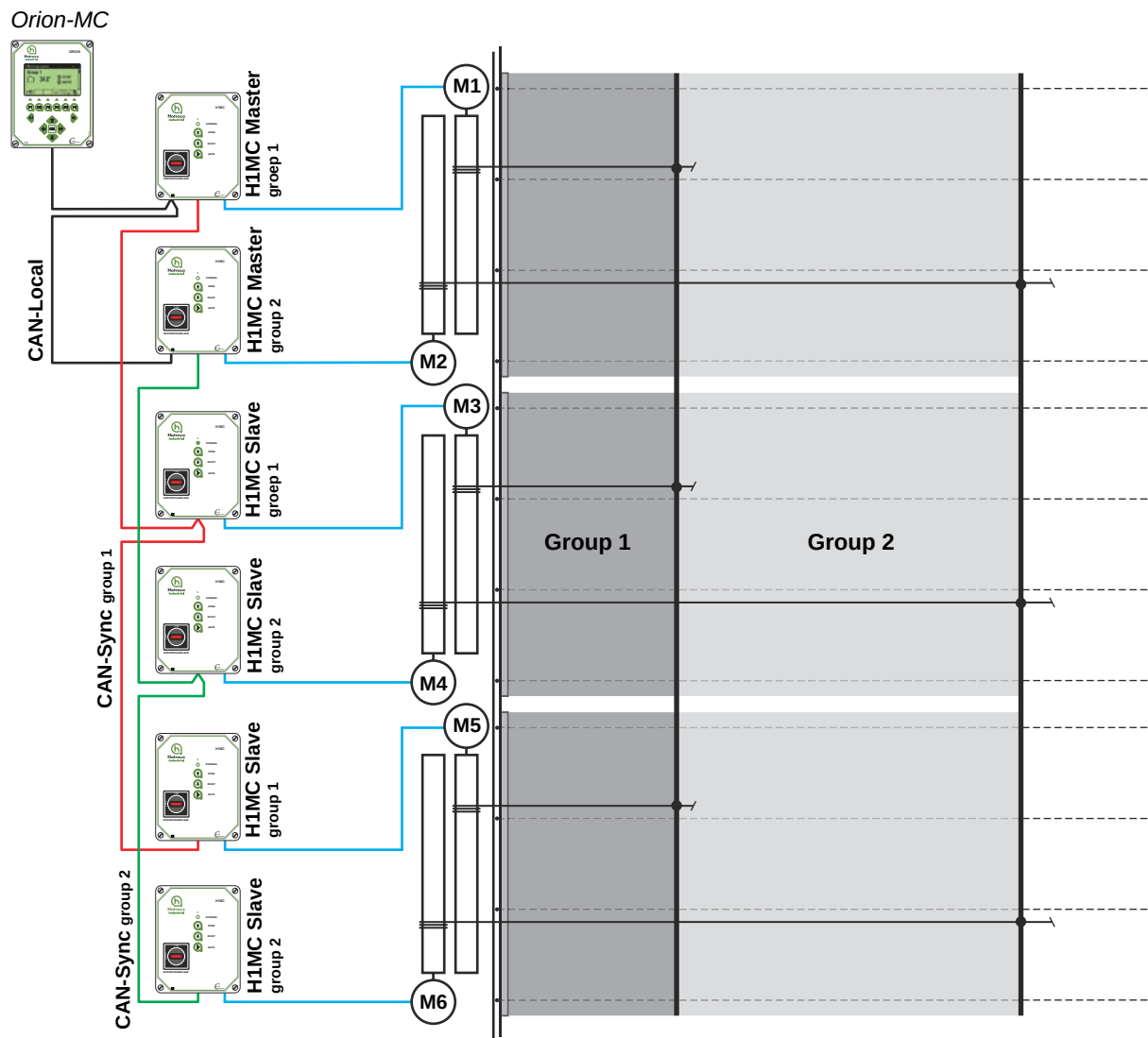
IO modules H2MC and H1MC (example)



IO module H2MC with frequency controller (example)



IO module H1MC master-slave link (example)



In this example each screen has a separate drive motor, and the screens are linked to each other mechanically. The motors M1, M3 and M5 drive group 1. The *H1MC master* controls motor M1 and communicates with the Orion-MC (via CAN Local). The other motors (M3 and M5) are controlled by an *H1MC slave*. The *H1MC slaves* communicate with the *H1MC master* via what is referred to as the CAN-Sync Data Bus (see also the H1MC manual). The same is applicable to group 2, which is driven by the motors M2, M4 and M6.

Opposing screens

Without master group

When the main controller has controlled the two screens to a position that would result in an overlap, then the screen closed last will go into the desired position. The other screen is then opened automatically. A small gap is maintained between the two screen sections.

Example G1= motors group 1, G2 = motors group 2

G2 is closed. When group G2 is close to the opposite group G1, this group will be opened. The virtual motor of G1 stays in its original position to make sure that the main controller gets the correct feedback.



When group G2 has reached the desired position, both screens will be stopped.



With an assigned master group

In the following example, group 1 is the master group. When the main controller controls the two screens to a position that would result in an overlap, then the screen of the master group will go to the desired position, irrespective of the position of the other screen. The other screen is then opened automatically, if necessary. A gap is maintained between the two screen sections.

Example G1= motors of group 1 (master), G2 = motors group 2

G2 is closed. When group G2 is close to the opposite group G1 (master group), then group G1 will stop. Group G1 keeps the set position. The virtual motor of group G2 continues to the desired position to make sure that the main controller gets the correct feedback.



Once G1 (master) is opened, G2 will go to the desired position.



When G1 (master) is opened, and this position would overlap with group G2, then G2 will be sent back.



Consecutive

Without master group

The Orion-MC makes sure that the screen closed last goes into the desired position. The position of the other screen is then changed automatically, without a folded screen getting squashed. A minimum distance is kept between the two folded screens while the screens are sliding.

Example G1= motors of group 1, G2 = motors of group 2

Group G1 is fully open. Group G2 is fully open, but is now closed by the main controller.



Group 1 will now move with group 2, whereby group G1 leads group G2 slightly.



A group is stopped once it reaches its end position.



Example G1= motors of group 1, G2 = motors of group 2

Group G1 is open, and group G2 is closed. When the controller sends the command to close group G1, the Orion-MC will send the command to open group G2, as a result of which group G will be closed automatically.



With an assigned master group

The screen of the master group will automatically go to the desired position, irrespective of the position of the other screen. A minimum distance is kept between the two folded screens while the screens are sliding.

G1= motors of group 1, G2 = motors of group 2 (master)

Group G1 is partly closed, and group G2 (master) is partly closed too.



When the main controller wants to close group G1 further, then group G2 would normally be sent back at the time group G1 has reached the end position.



However, because group G2 is the master group, this will not occur. The virtual motor of group 1 will go to the desired position.



Only when group G2 (master) is opened, then group G1 will be closed further by sliding group G2.

